
Reliable Causal Estimation under Latent Markov Confounding

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Abstract

Causal inference from dependent observational time series is complicated when an unobserved, persistent state jointly affects treatment assignment and outcomes. Standard Double Machine Learning (DML) controls for observed covariates through orthogonal residualization, but does not, by itself, resolve confounding induced by an imperfect proxy for latent state dynamics. We study causal estimation under latent Markov confounding and characterize how latent-state uncertainty and regime overlap propagate into causal estimation error. We develop a regime-aware orthogonal estimator that combines probabilistic state inference, temporally purged cross-fitting, and regularized inversion of the cross-regime score Jacobian. Our analysis separates nuisance-estimation error, latent-state recovery error, and variability induced by persistent regime occupation, and gives conditions under which the estimator approaches the corresponding oracle-state estimator as latent-state uncertainty decreases. We evaluate these properties using real-data-calibrated semi-synthetic time series constructed from high-frequency atmospheric observations across heterogeneous Indian cities, preserving persistence, seasonality, missingness, and regime overlap while keeping causal effects known. Across controlled variations in state persistence and observability, the experiments show how latent-state uncertainty predicts causal bias and when regime-aware estimation improves over standard and temporally blocked DML. Real sensor networks provide an observational stress test, revealing substantial differences in es-

timated effects across inferred atmospheric states. Together, the results show that reliable causal estimation with latent sequential confounding is governed jointly by latent-state uncertainty and downstream task conditioning, and provide diagnostics for when regime-aware estimation should be trusted or treated as unreliable.

1 INTRODUCTION

Sequential machine learning systems frequently operate in environments governed by persistent, partially observed context—such as unobserved physical, macroeconomic, or operational regimes. A widespread design assumption in such systems is that predictive confidence or calibration over the latent state is sufficient for downstream reliability. However, this assumption fails: *a model can be confident about a hidden state and still be unreliable for the downstream task because downstream task geometry governs failure*. When the downstream task is ill-conditioned—such as estimating causal effects under collinear treatments or weak overlap—even small posterior proxy error is catastrophically amplified, causing severe downstream failure despite seemingly confident latent representations. The same latent-state uncertainty can be informative for one task and nearly useless for another because the downstream information geometry differs.

Continuous-treatment causal inference in atmospheric systems provides a natural laboratory: estimating the causal effect of variation in the observed NO₂ exposure proxy on ambient PM_{2.5} (Dominici et al., 2014) (translating this observational exposure effect to an emissions policy requires structural transport assumptions). Observational time series in this domain violate two foundational assumptions: (1) **Persistent Latent Regime Confounding**: Planetary boundary-layer dynamics transition between discrete thermodynamic regimes (e.g., nocturnal inversion vs. convective clearance), jointly governing precursor dispersion (T_t) and ambient particulate concentration (Y_t). (2) **Imperfect Overlap & Dependence**: Latent regimes

are unobserved; auxiliary proxies Z_t exhibit overlap ($\varepsilon_\gamma > 0$), while autoregression breaks exchangeability (Chernozhukov et al., 2018), leaking errors across folds.

When standard Double Machine Learning (Chernozhukov et al., 2018) or tree-based plug-in estimators are applied naively to environmental time series without accounting for latent regimes, the resulting causal estimands can exhibit severe bias and, in our benchmark, sign inversions—falsely suggesting that precursor reductions exacerbate ambient pollution.

Related Work. We position our contribution across four neighboring literatures: (1) **Double Machine Learning & Orthogonal Inference:** Chernozhukov et al. (2018, 2022); Foster and Syrgkanis (2023); Nie and Wager (2021); Kennedy (2023) achieve $\sqrt{N}$ -rate inference under unconfoundedness given X_t . (2) **Dependent & Time-Series Causal Inference:** Ciganovic et al. (2026); Semenova (2023); Lewis and Syrgkanis (2020); Bai (2009); Bonhomme and Manresa (2015); Hamilton (1989) relax exchangeability under mixing, but do not address latent Markov confounding. (3) **Latent Markov & Proximal Inference:** Clouth et al. (2024); Huang et al. (2023); Miao et al. (2018); Tchetgen Tchetgen et al. (2020); Mastouri et al. (2021); Cappé et al. (2005) use proxies for unmeasured confounders, focusing on asymptotic identification rather than downstream finite-sample score conditioning. (4) **Weak Overlap & Selective Reliability:** D’Amour et al. (2021); Shalit et al. (2017); Crump et al. (2009); Guo et al. (2017); Geifman and El-Yaniv (2017); Sankararaman et al. (2023) address overlap and abstention, without linking latent uncertainty to coupled score conditioning.

Contributions. We separate latent-state uncertainty from task conditioning and show how their interaction controls downstream degradation:

- **Decoupling Context Uncertainty from Task Conditioning:** We prove that downstream estimation suffers two distinct degradation mechanisms: imperfect latent recovery (ε_γ) and weak task conditioning ($\lambda_{\min}(\mathbf{J})$), showing that predictive state confidence does not imply downstream task reliability (Theorems 3–4).
- **Principled Difficulty Measure & Observable Bridge:** We formulate causal difficulty $\mathcal{D}_{\text{causal}} = \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}$ and establish rolling entropy $\mathcal{D}_{\text{operational}}(t) = \frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J}_t)}$ as an observable multiplicative surrogate under two-sided spectral comparability (Proposition 5).
- **Sequential Overlap-Aware Estimation (OR-DML):** We develop spectrally regularized OR-DML with purged block cross-fitting, proving leading-order risk bounds and establishing cen-

tered asymptotic normality under dependent data (Theorems 6–7).

- **Representation Zoo & Multi-Domain Empirical Evidence:** Across controlled multi-world benchmarks, $> 14,000$ hours of Indian sensor data, and a synthetic financial stress test, we show that deep sequence encoders (GRU, Transformer, SSM) with high state $F_1 \approx 0.85$ fail downstream reliability ($\text{AUC} \approx 0.52$) due to poor calibration, whereas well-calibrated HMM causal filtering coupled with task geometry achieves $\text{AUC} = \mathbf{0.6153}$.

2 STRUCTURAL CAUSAL MODEL & IDENTIFICATION

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Consider a discrete-time longitudinal process observed at equidistant intervals $t \in \{1, \dots, N\}$.

2.1 Structural Equations

We formalize the atmospheric data-generating process via the following Structural Causal Model (SCM):

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad S_t \in \{1, \dots, K\}, \quad (1)$$

$$Z_t = \mu_Z(S_t) + \xi_t, \quad (2)$$

$$X_t = \phi(X_{t-1}, S_t) + \epsilon_t, \quad (3)$$

$$T_t = m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \quad (4)$$

$$Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad (5)$$

with $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$, where:

- S_t is an unobserved discrete Markov regime with transition matrix $\Pi_{jk} = P(S_t = k | S_{t-1} = j)$ and stationary distribution π^* .
- $Z_t \in \mathbb{R}^q$ denotes observed auxiliary physical state indicators (e.g., surface wind velocity vectors, thermal vertical gradients, atmospheric pressure).
- $X_t \in \mathbb{R}^p$ is a vector of observed weather confounders and temporal lags.
- $T_t \in \mathbb{R}$ is the continuous treatment (precursor emission proxy, e.g., NO_2).
- $Y_t \in \mathbb{R}$ is the continuous outcome (ambient pollutant, e.g., $\text{PM}_{2.5}$).
- $\theta(S_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k)$ represents the regime-specific causal effect.

2.2 Identification Under Latent Regimes

Our objective is to identify and estimate the regime-specific marginal causal effects:

$$\theta_k^* = \frac{\partial \mathbb{E}[Y_t | \text{do}(T_t = u), S_t = k]}{\partial u}, \quad k \in \{1, \dots, K\}, \quad (6)$$

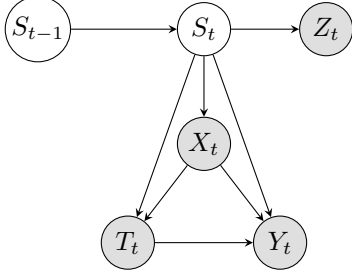


Figure 1: Temporal Causal Graph. Latent transitions $S_{t-1} \rightarrow S_t$ govern persistent regime dynamics; S_t acts as a common cause for proxy Z_t , confounders X_t , treatment T_t , and outcome Y_t . Omitting S_t induces omitted regime bias.

as well as the stationary Average Treatment Effect (ATE):

$$\theta_{\text{ATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^*. \quad (7)$$

Assumption 1 (Regime-Conditional Unconfoundedness & Innovation Exogeneity). Conditional on observed covariates X_t , latent regime S_t , and proxy history $Z_{1:N}$, potential outcomes are unconfounded ($Y_t(u) \perp T_t \mid (X_t, S_t, Z_{1:N}), \forall u$) and structural innovations have zero mean:

$$\mathbb{E}[U_t \mid X_t, S_t, T_t, Z_{1:N}] = 0. \quad (8)$$

Assumption 2 (Overlap & Positivity). There exist constants $c_1, c_2 > 0$ such that for all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t \mid X_t, S_t = k) \leq c_2. \quad (9)$$

Assumption 3 (Latent Proxy Sufficiency). The auxiliary state proxy sequence $Z_{1:N}$ satisfies latent proxy sufficiency: $P(S_t = k \mid X_t, T_t, Z_{1:N}) = P(S_t = k \mid Z_{1:N}) \equiv \gamma_{tk}$ (or $P(S_t = k \mid Z_{1:t})$ for causal forward filtering) almost surely for all $k \in \{1, \dots, K\}$. *Discussion.* That is, $Z_{1:N}$ screens off the latent state S_t from (X_t, T_t) , framing it as a proxy-sufficiency identification restriction. Combined with Assumption 1, this guarantees $\mathbb{E}[Y_t \mid X_t, T_t, Z_{1:N}] = \sum_{k=1}^K \gamma_{tk} [\theta_k^* T_t + g_k(X_t)]$. This identifying restriction requires $Z_{1:N}$ to be sufficiently informative that conditioning on (X_t, T_t) provides no additional regime information beyond $Z_{1:N}$.

Assumption 4 (Nuisance Estimation and Mixing Rates). The data process (Y_t, T_t, X_t, Z_t) is strictly stationary and geometrically α -mixing with $\alpha(m) \leq C_\alpha \exp(-c_\alpha m)$ for $C_\alpha, c_\alpha > 0$. Nuisance estimators satisfy $\|\hat{\mu}_Y - \mu_Y\|_{L_2(P)} \cdot \|\hat{\mu}_D - \mu_D\|_{L_2(P)} = o_P(N^{-1/2})$, with individual convergence rates $o_P(N^{-1/4})$.

3 FAILURE OF STANDARD DML: THE OMITTED REGIME BIAS

Intuition. When unobserved regimes S_t jointly shift treatment and outcome baselines, partialling out observed controls X_t cannot purge latent regime variation. Instead, by reducing residual treatment variance in the denominator without eliminating the latent covariance in the numerator, standard cross-sectional residualization actively amplifies the omitted confounding.

Theorem 1 (Decomposition of Omitted Regime Bias). *Suppose the data are generated according to SCM (1)–(5). Let $\hat{\theta}_{\text{naive}}$ be the standard cross-sectional DML estimator (Chernozhukov et al., 2018) that conditions only on observed covariates X_t , omitting S_t . Define the naive cross-sectional residuals $\tilde{T}_t = T_t - \mathbb{E}[T_t \mid X_t]$ and $\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t]$.*

Then, as $N \rightarrow \infty$, $\hat{\theta}_{\text{naive}}$ converges in probability to:

$$\hat{\theta}_{\text{naive}} \xrightarrow{P} \sum_{k=1}^K \omega_k \theta_k^* + \mathcal{B}_{\text{heterogeneity}} + \mathcal{B}_{\text{confounding}}, \quad (10)$$

where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t=k)\tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$, and the bias components are:

$$\mathcal{B}_{\text{heterogeneity}} = \frac{\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (11)$$

$$\mathcal{B}_{\text{confounding}} = \frac{\sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (12)$$

with $\pi_k(X_t) \equiv P(S_t = k \mid X_t)$, $\bar{m}(X_t) \equiv \mathbb{E}[T_t \mid X_t]$, $\Delta m_k(X_t) \equiv \mathbb{E}[T_t \mid X_t, S_t = k] - \bar{m}(X_t)$, and $\Delta g_k(X_t) \equiv g(X_t, k) - \mathbb{E}[g(X_t, S_t) \mid X_t]$ (complete algebraic proof in Appendix D).

Corollary 2 (Sign Inversion & Bias Amplification). *When latent stagnation induces strong positive confounding ($\Delta m_k(X_t) \Delta g_k(X_t) \gg 0$), but residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ is attenuated due to conditioning on X_t , the omitted confounding bias $\mathcal{B}_{\text{confounding}}$ undergoes severe bias amplification. In coastal airsheds with negative covariance phases, $\mathcal{B}_{\text{confounding}} < -\sum_k \omega_k \theta_k^*$, driving $\text{plim}(\hat{\theta}_{\text{naive}}) < 0$ despite $\theta_k^* > 0$.*

Interpretation. Standard DML cannot eliminate omitted regime confounding: conditioning on observed controls X_t shrinks residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ faster than it reduces latent covariance, actively amplifying bias.

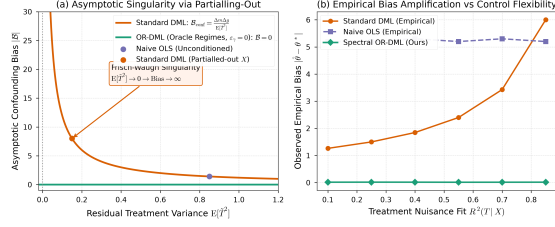


Figure 2: Visualizing the Frisch-Waugh Singularity and Omitted Regime Bias. (a) As residual treatment variance $\mathbb{E}[\tilde{T}_t^2] \rightarrow 0$ via partialling-out X_t , omitted regime bias $\mathcal{B}_{\text{confounding}}$ explodes hyperbolically in Standard DML, while OR-DML (Oracle Regimes, $\varepsilon_\gamma = 0$) remains strictly unbiased ($\mathcal{B} = 0$). (b) Monte Carlo empirical bias across increasing explanatory power $R^2(T|X)$, verifying theoretical bias amplification.

4 METHODOLOGY: OVERLAP-AWARE REGIME DML

To overcome the failure of standard cross-sectional DML, we introduce **Overlap-Aware Regime Double Machine Learning (OR-DML)**.

4.1 Latent Regime Formulation & Observability Diagnostics

Let $Z_t \in \mathbb{R}^q$ denote exogenous contemporaneous meteorological covariates assumed unaffected by the exposure proxy. We model the latent atmospheric state $S_t \in \{1, \dots, K\}$ via a Hidden Markov Model with transition matrix $\mathbf{\Pi}$ and emission parameters ϕ . Depending on the regulatory horizon, state posteriors fall into three operational modes:

1. **Retrospective Smoothed Posteriors (Offline):** $\gamma_{tk}^{\text{smooth}} \equiv P(S_t = k | Z_{1:N})$ for retrospective environmental epidemiology and structural causal auditing;
2. **Training-Fitted Forward Filtered Posteriors:** $\gamma_{tk}^{\text{filter}} \equiv P(S_t = k | Z_{1:t}, \hat{\phi})$ where dynamics $\hat{\phi}$ are calibrated on a historical window strictly preceding evaluation ($\mathcal{T}_{\text{cal}} < \mathcal{T}_{\text{eval}}$), with state updates proceeding strictly causally on $Z_{1:t}$; or
3. **Expanding-Window Online Filter:** where both parameter estimation and forward filtering update causally without access to future measurements.

Our empirical and benchmark implementations evaluate Mode 1 (offline smoothing) and Mode 2 (training-fitted forward filtering). We define the mean posterior entropy as a direct diagnostic of latent state observ-

ability:

$$\bar{H}(\gamma) \equiv -\frac{1}{N} \sum_{t=1}^N \sum_{k=1}^K \gamma_{tk} \ln(\gamma_{tk} + \epsilon). \quad (13)$$

A Priori Diagnostics: Standardized separation $\Delta_Z < 0.5$ or entropy $\bar{H} \rightarrow \ln K$ signals severe proxy uncertainty and triggers a weak-identification diagnostic warning (rules in Appendix B.3), distinguishing structural non-identification ($\Delta_Z = 0$, Theorem 3) from finite-sample proxy dilution.

4.2 State-Weighted Coupled System & Spectral Regularization

To avoid heuristic decoupled regressions that suffer cross-talk bias under regime overlap, OR-DML estimates regime-specific conditional expectations via state-weighted regressions targeting population expectations $m_k^\gamma(x) \equiv \frac{\mathbb{E}[\gamma_{tk} T_t | X_t = x]}{\mathbb{E}[\gamma_{tk} | X_t = x]}$ and $\mu_k^\gamma(x) \equiv \frac{\mathbb{E}[\gamma_{tk} Y_t | X_t = x]}{\mathbb{E}[\gamma_{tk} | X_t = x]}$:

$$\hat{m}_k(X) = \arg \min_m \sum_{t \in \mathcal{T}_b} \gamma_{tk} (T_t - m(X_t))^2, \quad (14)$$

$$\hat{\mu}_k(X) = \arg \min_\mu \sum_{t \in \mathcal{T}_b} \gamma_{tk} (Y_t - \mu(X_t))^2. \quad (15)$$

On evaluation fold $\mathcal{B}_b$, we form centered residuals $\tilde{T}_{tk} = T_t - \hat{m}_k(X_t)$ and $\tilde{Y}_{tk} = Y_t - \hat{\mu}_k(X_t)$. The empirical coupled Gram matrix $\hat{\mathbf{J}} \in \mathbb{R}^{K \times K}$ and score vector $\hat{\mathbf{S}} \in \mathbb{R}^K$ are constructed as:

$$\hat{J}_{jk} = \frac{1}{N} \sum_{t=1}^N \gamma_{tj} \gamma_{tk} \tilde{T}_{tj} \tilde{T}_{tk}, \quad \hat{S}_k = \frac{1}{N} \sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk} \tilde{Y}_{tk}. \quad (16)$$

When regime separation is weak or treatments are collinear, $\lambda_{\min}(\hat{\mathbf{J}}) \rightarrow 0$, rendering the unregularized solve $\hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ ill-conditioned. OR-DML introduces the **Spectrally Regularized Estimator**:

$$\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}, \quad (17)$$

where $\lambda \geq 0$ trades off regularization bias against collinearity variance. *Scalability & Structural Abstinence:* End-to-end complexity is $\mathcal{O}(BK \cdot C_{\text{nuisance}} + NK^2 + K^3)$, with Gram construction taking $\mathcal{O}(NK^2)$ and Cholesky solve taking $\mathcal{O}(K^3)$ (Appendix C.2); when data lack identifying variation entirely ($\kappa(\hat{\mathbf{J}}) > 10^4$ or $\text{CI}(\lambda_{\min})$ covers 0), systems should diagnose weak identification and abstain rather than masking near-singularity with excessive shrinkage. We distinguish three causal estimand targets: (i) the true stationary population ATE $\theta_{\text{PATE}}^* = \pi^{*\top} \theta^*$, (ii) the estimated stationary ATE $\hat{\theta}_{\text{PATE}} = \hat{\pi}(\hat{\mathbf{A}})^\top \hat{\theta}_\lambda$, and (iii) the empirical sample occupation ATE $\hat{\theta}_{\text{SATE}} = \bar{\gamma}_N^\top \hat{\theta}_\lambda$, where $\bar{\gamma}_N \equiv \frac{1}{N} \sum_{t=1}^N \gamma_t$.

Algorithm 1 Overlap-Aware Regime DML (OR-DML)

Require: Series $\{Y_t, T_t, X_t, Z_t\}_{t=1}^N$, blocks B , embargo τ^* , penalty $\lambda \geq 0$.

- 1: Infer posteriors γ_t (Mode 1 smoothing or Mode 2 forward filtering).
 - 2: Cross-fit state-weighted nuisances $\hat{m}_k, \hat{\mu}_k$ with embargo buffer τ^* .
 - 3: Form centered residuals $\tilde{T}_{tk} = T_t - \hat{m}_k(X_t)$ and $\tilde{Y}_{tk} = Y_t - \hat{\mu}_k(X_t)$.
 - 4: Form Gram $\hat{\mathbf{J}} = \frac{1}{N} \sum_t \gamma_t \gamma_t^\top \odot \tilde{\mathbf{T}}_t \tilde{\mathbf{T}}_t^\top$ and score $\hat{\mathbf{S}} = \frac{1}{N} \sum_t \gamma_t \odot \tilde{\mathbf{T}}_t \odot \tilde{\mathbf{Y}}_t$.
 - 5: Compute $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ and diagnostics $(\lambda_{\min}, \kappa, \bar{H})$.
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4.3 Purged Block Cross-Fitting with Bartlett Embargo

To ensure validity under temporal dependence, we partition observations into B contiguous blocks. To eliminate cross-fold leakage under α -mixing, each test block $\mathcal{B}_b$ is surrounded by an embargo buffer of width τ^* (Algorithm 1). We determine τ^* from the data via the Bartlett significance threshold:

$$\tau^* = \max \left\{ h \geq 1 : \max_j |\hat{\rho}_j(h)| \geq \frac{1.96}{\sqrt{N}} \right\} + 1. \quad (18)$$

5 THEORETICAL ANALYSIS

We formalize three central results: (1) **Non-Identification:** latent context can induce unidentifiability; (2) **Graceful Degradation:** proxy error scales as $\varepsilon_\gamma / \lambda_{\min}(\mathbf{J})$; and (3) **Observable Difficulty & Regularization:** task conditioning yields an observable surrogate for risk control.

Intuition. If auxiliary proxies Z_t contain zero information distinguishing latent regimes, different underlying causal structures generate observationally indistinguishable joint distributions, rendering point identification impossible regardless of sample size.

Theorem 3 (Non-Identification Under Unconstrained Overlap). *Suppose the latent state is unobserved, and auxiliary proxies Z_t satisfy $P(Z_t \mid S_t = j) = P(Z_t \mid S_t = k)$ for all j, k . Then the regime-specific causal effect vector $\boldsymbol{\theta}^* = (\theta_1^*, \dots, \theta_K^*)^\top$ is not point-identified from the joint distribution of observables (Y_t, T_t, X_t, Z_t) : there exist structural causal models P_1 and P_2 satisfying Assumptions 1–3 with identical joint observable laws $P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N})$ for all $N \geq 1$, yet possessing strictly distinct, non-permutation causal effect vectors $\boldsymbol{\theta}^{(1)} \neq \boldsymbol{\theta}^{(2)}$.*

Interpretation. No estimator can recover regime-specific effects when the proxy provides no state information; this is a fundamental limitation of the observable information structure, not of optimization or sample size (proof in Appendix A).

Intuition. Downstream estimation error has two qualitatively different sources: posterior proxy error ε_γ from imperfect state inference, and causal ill-conditioning $\lambda_{\min}(\mathbf{J})$ from collinear residual treatment variation. Theorem 4 proves that these mechanisms enter multiplicatively.

Theorem 4 (Graceful Degradation Error Bound). *Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator, and let $\gamma_t \in \Delta^K$ be an estimated posterior state proxy with average L_1 proxy error $\varepsilon_\gamma \equiv \mathbb{E} \|\gamma_t - \mathbf{H}_t\|_1$. Suppose $\|\boldsymbol{\theta}^*\|_2 \leq M_\theta < \infty$, $\sup_t |\tilde{T}_{tk}^\gamma| \leq M_T < \infty$ almost surely, and regime functions satisfy $\max_{j,k} |g(x, j) - g(x, k)| \leq L_g < \infty$ and $\max_{j,k} |m_j^\gamma(x) - m_k^\gamma(x)| \leq L_m < \infty$. Let $\mathbf{J} \in \mathbb{R}^{K \times K}$ with $J_{jk} \equiv \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma]$ and $\mathbf{S} \in \mathbb{R}^K$ with $S_k \equiv \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma]$ denote the population coupled Gram matrix and score vector targeting implemented proxy-weighted nuisances $(m_k^\gamma, \mu_k^\gamma)$. If $\lambda_{\min}(\mathbf{J}) > 0$, the population estimand $\boldsymbol{\theta}_\gamma \equiv \mathbf{J}^{-1} \mathbf{S}$ satisfies:*

$$\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}, \quad (19)$$

and the unregularized cross-fitted estimator $\hat{\boldsymbol{\theta}}_\gamma = \hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ satisfies $\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2})$, where $C = C_S + M_\theta C_J < \infty$ depends on bounded moments and envelope constants (Appendix B).

Interpretation. Better state recovery alone is not enough: when downstream task geometry is ill-conditioned ($\lambda_{\min}(\mathbf{J}) \approx 0$), even tiny proxy errors are catastrophically amplified into large causal bias.

Theorem 4 replaces heuristic orthogonality claims with a rigorous bound connecting *posterior proxy error* ε_γ and *Jacobian conditioning* $\lambda_{\min}(\mathbf{J})$. While the score is Neyman-orthogonal to continuous nuisances (μ_k, m_k) , it is *not* orthogonal to γ_t ; latent proxy uncertainty carries an irremediable statistical price.

The Difficulty Frontier and Operational Surrogate Bridge.

Theorem 4 demonstrates that asymptotic bias scales with theoretical ratio $\mathcal{D}_{\text{causal}} \equiv \varepsilon_\gamma / \lambda_{\min}(\mathbf{J})$, where $\varepsilon_\gamma \equiv \mathbb{E} \|\gamma_t - \mathbf{H}_t\|_1$ is evaluated in simulation via $\hat{\varepsilon}_{\gamma, N} \equiv \frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbf{H}_t\|_1$. Because true states are unobserved in deployment, we bridge unobservable $\mathcal{D}_{\text{causal}}$ to computable operational index $\mathcal{D}_{\text{operational}}(t) \equiv H(\gamma_t) / \lambda_{\min}(\mathbf{J}_t)$ via the following result under spectral comparability:

Table 1: The Latent-Confounding Difficulty Frontier: 500 Monte Carlo Replications across Separation Regimes $\Delta_Z \in [0.2, 4.0]$ ($N = 1, 200$, True $\theta_{\text{ATE}}^* = 1.5278$). Nominal coverage target is 95.0% for Sample ATE (SATE) and Population ATE (PATE). Complete 10-column evaluation with $\kappa(\mathbf{J})$ and $\hat{\varepsilon}_\gamma$ reported in Table 7 (Appendix H).

Δ_Z	Estimator	Abs Bias	RMSE	95% Cov (S/P)	$\lambda_{\min}(\mathbf{J})$
0.2 (Severe Overlap)	Oracle DML	0.0558	0.0703	96.0% / 94.0%	0.4485
	Standard DML	8.4556	8.4562	0.0% / 0.0%	4.9339
	Unreg. Coupled DML	8.6066	8.6077	0.0% / 0.0%	0.6495
	Spectral OR-DML (Ours)	8.5958	8.5969	0.0% / 0.0%	0.6495
0.5 (Weak Separation)	Oracle DML	0.0552	0.0713	92.0% / 95.0%	0.4480
	Standard DML	8.4287	8.4296	0.0% / 0.0%	4.9560
	Unreg. Coupled DML	8.8095	8.8138	0.0% / 0.0%	0.7680
	Spectral OR-DML (Ours)	8.7974	8.8017	0.0% / 0.0%	0.7680
1.0 (Moderate)	Oracle DML	0.0572	0.0735	92.0% / 92.0%	0.4515
	Standard DML	8.4331	8.4340	0.0% / 0.0%	4.9535
	Unreg. Coupled DML	4.2555	4.3823	0.0% / 0.0%	0.5294
	Spectral OR-DML (Ours)	4.2481	4.3757	0.0% / 0.0%	0.5294
2.0 (Clean Separation)	Oracle DML	0.0521	0.0658	93.0% / 97.0%	0.4536
	Standard DML	8.4318	8.4327	0.0% / 0.0%	4.9539
	Unreg. Coupled DML	0.5020	1.9036	89.0% / 93.0%	0.4321
	Spectral OR-DML (Ours)	0.5012	1.9048	87.0% / 92.0%	0.4321
	Filtered OR-DML (Ours)	0.5203	1.9062	87.0% / 90.0%	0.4329
4.0 (Isolated Regimes)	Oracle DML	0.0540	0.0670	96.0% / 95.0%	0.4394
	Standard DML	8.4509	8.4518	0.0% / 0.0%	4.9123
	Unreg. Coupled DML	0.2236	1.2113	94.0% / 93.0%	0.4299
	Spectral OR-DML (Ours)	0.2233	1.2102	94.0% / 93.0%	0.4299
	Filtered OR-DML (Ours)	0.2233	1.2100	94.0% / 93.0%	0.4299

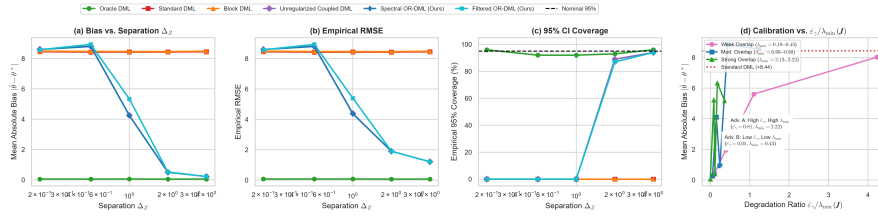


Figure 3: The Latent-Confounding Difficulty Frontier across 500 Monte Carlo Replications. (a) Absolute causal bias $|\hat{\theta} - \theta^*|$ vs. regime separation Δ_Z . (b) Empirical RMSE. (c) Empirical 95% CI coverage vs. nominal 95% target. (d) Factorial calibration: Absolute bias $|\hat{\theta} - \theta^*|$ vs. theoretical degradation ratio $\bar{\varepsilon}_\gamma/\lambda_{\min}(\mathbf{J})$ (Theorem 4) across 15 factorial grid cells, validating the degradation scaling and highlighting Adversarial Corners A (high $\bar{\varepsilon}_\gamma$, healthy $\lambda_{\min}$) and B (low $\bar{\varepsilon}_\gamma$, weak $\lambda_{\min}$).

Assumption 5 (Two-Sided Spectral Comparability). There exist constants $0 < \underline{c}_J \leq \bar{c}_J < \infty$ such that rolling Gram matrices $\mathbf{J}_t$ satisfy $\underline{c}_J \leq \frac{\lambda_{\min}(\mathbf{J}_t)}{\lambda_{\min}(\mathbf{J})} \leq \bar{c}_J$ almost surely for all evaluation windows $t \geq w$.

Proposition 5 (Surrogate Entropy-Difficulty Bridge). Under correct posterior specification (Bayesian calibration) in binary regimes ($K = 2$), for causal forward-filtered posteriors $\gamma_t = P(S_t \in \cdot \mid \mathcal{F}_t)$ with filtration $\mathcal{F}_t = \sigma(Z_{1:t})$, the posterior error satisfies $\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t] = 2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t)$, where $C_K \leq 1/\ln 2 \approx 1.4427$. We define the difficulty hierarchy: $\mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}$, $\mathcal{D}_{\text{global}}^H(t) \equiv \frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J})}$, and $\mathcal{D}_{\text{operational}}(t) \equiv \frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J}_t)}$. (i) Causal difficulty is unconditionally bounded in population by global entropy: $\mathcal{D}_{\text{causal}} \leq C_K \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]$. (ii) Under Assumption 5,

the deployable index satisfies:

$$\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E}[\mathcal{D}_{\text{operational}}(t)] \leq \frac{C_K \bar{c}_J}{\underline{c}_J} \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]. \quad (20)$$

In simulation benchmarks where H_t is observed, $\hat{\mathcal{D}}_{\text{causal},N} \equiv \hat{\varepsilon}_{\gamma,N}/\lambda_{\min}(\mathbf{J})$ converges to $\mathcal{D}_{\text{causal}}$ with $\mathcal{O}_P(N^{-1/2})$ error for population posteriors $\gamma_t(\Lambda^*)$ (preserved under $\sqrt{N}$ -consistent $\hat{\Lambda}$).

Interpretation. Proposition 5 grounds deployable monitoring by turning an unobservable theoretical difficulty into an observable surrogate upper bound, establishing that posterior entropy bounds proxy error under Bayesian calibration.

Intuition. When treatments are collinear or regimes overlap, $\lambda_{\min}(\mathbf{J}) \rightarrow 0$ causes the unregularized covariance to blow up. Introducing a spectral ridge penalty $\lambda > 0$ bounds the effective condition number, trading finite-sample regularization bias for bounded estima-

tion variance.

Theorem 6 (Leading-Order Risk Bound under Long-Run Covariance Approximation). *Let $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ be the spectrally regularized OR-DML estimator, and let $\theta_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. Under Theorem 4, the regularized estimand satisfies the exact deterministic identity $\theta_\lambda - \theta^* = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda \theta^*)$, where $\|\mathbf{b}_\gamma\|_2 \leq C\varepsilon_\gamma$. Furthermore, under long-run score covariance $\Omega = \lim_{N \rightarrow \infty} \text{Var}(N^{-1/2} \sum_t \Psi_t)$, expected squared estimation error satisfies the leading-order risk bound:*

$$\mathbb{E} \|\hat{\theta}_\lambda - \theta^*\|_2^2 \leq \frac{(C\varepsilon_\gamma + \lambda M_\theta)^2 + N^{-1} \text{Tr}(\Omega)}{(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + \mathcal{O}(N^{-1}), \quad (21)$$

accounting for stochastic expansion $\mathbb{E} \|\hat{\theta}_\lambda - \theta_\lambda\|_2^2 = \frac{1}{N} \text{Tr}((\mathbf{J} + \lambda \mathbf{I})^{-1} \Omega (\mathbf{J} + \lambda \mathbf{I})^{-1}) + o(N^{-1})$ and deterministic shrinkage (Appendix C), sharpening to $\mathcal{O}(N^{-3/2})$ under $\varepsilon_\gamma + \lambda = \mathcal{O}(N^{-1/2})$. In practice, plug-in surrogate $\hat{R}(\lambda)$ yields $\hat{\lambda}_{\text{emp}}^* = \arg \min_{\lambda \geq 0} \hat{R}(\lambda)$ (Appendix C.1, where $\lambda = 0.10$ achieves an empirical $23.0 \times$ risk reduction in ill-conditioned benchmarks with 85.0% coverage), distinct from vanishing penalty $\lambda = o(N^{-1/2})$ for centered inference.

Interpretation. Regularization provides finite-sample variance control when overlap is weak, but cannot manufacture causal information that is absent from the proxy distribution.

Theorem 7 (Asymptotic Normality Under Dependent Purged Cross-Fitting). *Under Assumptions 1–4, with embargo buffer $\tau^* \geq C \log N$, the regularized OR-DML estimator under training-fitted forward filtering satisfies:*

$$\sqrt{N} (\hat{\theta}_\lambda - \theta^* - \text{Bias}(\lambda)) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_\lambda), \quad (22)$$

where $\Sigma_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \Omega (\mathbf{J} + \lambda \mathbf{I})^{-1}$ (with retrospective smoothing providing an offline descriptive counterpart). When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, recovering centered asymptotic normality.

Proposition 8 (Joint Asymptotic Distribution for Sample and Population ATE). *The sample ATE $\hat{\theta}_{\text{SATE}} = \bar{\gamma}^\top \hat{\theta}_\lambda$ has conditional variance $\sigma_{\text{SATE}}^2 = \bar{\gamma}^\top \Sigma_\lambda \bar{\gamma}$, where Markov persistence ($\rho \approx 0.88$) inflates the long-run variance of sample state occupation $\bar{\gamma}_N$ by $\frac{1+\rho}{1-\rho} \approx 15.7 \times$ relative to i.i.d. transitions. The population ATE $\hat{\theta}_{\text{PATE}} = \hat{\pi}(\hat{\Lambda})^\top \hat{\theta}_\lambda$ satisfies the joint delta expansion: $\sqrt{N}(\hat{\theta}_{\text{PATE}} - \theta_{\text{PATE}}^*) = \pi^{*\top} \sqrt{N}(\hat{\theta}_\lambda - \theta^*) + \theta^{*\top} \sqrt{N}(\hat{\pi} - \pi^*) + o_P(1)$, yielding asymptotic variance:*

$$\sigma_{\text{PATE}}^2 = \pi^{*\top} \Sigma_\lambda \pi^* + \theta^{*\top} \Sigma_\pi \theta^* + 2\pi^{*\top} \Sigma_{\theta\pi} \theta^*, \quad (23)$$

where $\Sigma_\pi = \nabla_\Lambda \pi^* \mathcal{I}_\Lambda^{-1} (\nabla_\Lambda \pi^*)^\top$ represents the parametric variance of the estimated stationary distribu-

tion derived from sequence-level HMM influence function $\phi_{\Lambda,t}$, and $\Sigma_{\theta\pi} = \text{LRCov}(\phi_{\theta,t}, \phi_{\Lambda,t})$. When $\text{LRCov}(\phi_\theta, \phi_\Lambda) = \mathbf{0}$, $\Sigma_{\theta\pi} = \mathbf{0}$; in general, $\Sigma_{\theta\pi}$ is consistently estimated via the stacked joint HAC sandwich estimator.

6 THE LATENT-CONFOUNDING DIFFICULTY FRONTIER

To test the theoretical predictions of Theorems 3–6, we conduct a comprehensive Monte Carlo evaluation across 500 replications, charting the causal estimation error across the **Latent-Confounding Difficulty Frontier**.

Experimental Protocol. *DGPs & Baselines:* 500 Monte Carlo draws across $\Delta_Z \in \{0.2, \dots, 4.0\}$ ($N = 1, 200$, $\theta_{\text{ATE}}^* = 1.5278$) comparing Oracle, Standard, and OR-DML. *Representations & Shifts:* HMM, GRU, Transformer, and SSM under clean and compound shifts. *Reliability Target:* Predicting downstream squared loss via difficulty $\mathcal{D}_t$ across 5 seeds with MBB. *Real-World Test:* $> 14,000$ hourly observations across four megacities.

Difficulty Frontier Findings. Table 1 and Figure 3 reveal three phenomena:

1. **Naive Confounding & Severe Overlap Abstention:** Standard DML exhibits constant omitted-regime bias (+8.44, RMSE 8.45) and 0.0% coverage across all Δ_Z (Theorem 1). Under severe overlap ($\Delta_Z \leq 0.5$), coupled estimation with noisy posteriors ($\hat{\varepsilon}_\gamma = 0.83$) yields biases (8.60–8.80) matching naive pooling. By Theorem 3, this regime is unidentifiable; high operational difficulty $\mathcal{D}_t$ acts as an explicit *abstention trigger*.
2. **Graceful Degradation & Conditioning Deception:** By Theorem 4, OR-DML separates proxy error $\varepsilon_\gamma / \lambda_{\min}(\mathbf{J})$ from estimation error $\mathcal{O}_P(N^{-1/2})$. At $\Delta_Z \leq 0.5$, uniform posteriors symmetrically scale Gram entries, producing deceptively small $\kappa(\mathbf{J}) \approx 3.2$ –6.7 despite severe error ($\hat{\varepsilon}_\gamma = 0.83$). As regimes separate ($\Delta_Z = 2.0$), eigenvalues split, exposing true collinearity ($\kappa = 690.83$) where spectral shrinkage prevents variance explosion. Across $\Delta_Z \in [0.2, 4.0]$, bias drops by 97.4% (8.60 $\rightarrow$ 0.22), with coverage reaching 93.0%.
3. **Filtering vs. Smoothing:** Causal forward filtering (γ_t^{filter}) tracks retrospective smoothing (γ_t^{smooth}) closely across separated regimes (at $\Delta_Z = 2.0$, bias 0.5203 vs. 0.5012; at $\Delta_Z = 4.0$, identical at 0.2233), incurring virtually zero efficiency loss once dynamics are informed by physical history.

Table 2: Empirical Regime-Specific Exposure Effects across Four Major Indian Megacities ($> 14,000$ Hourly Observations, Feb 2025–Jun 2026). Treatment $T_t = \text{NO}_2$ Exposure Proxy, Outcome $Y_t = \text{PM}_{2.5}$. Complete regime decompositions and sensitivity profiles are reported in Appendix G (Tables 4–6).

City	Estimator	Regime	Effect θ	Std Error	95% CI	p-value	$\lambda_{\min}(J)$ [K]
Delhi ($N = 4,640$)	Spectral OR-DML	Regime 1	+0.7038	0.0513	[+0.6032, +0.8044]	< 0.0001	111.8 [1.58]
Mumbai ($N = 4,180$)	Spectral OR-DML	Regime 2	+3.3036	1.4800	[+0.4027, +6.2045]	0.0256	0.32 [1.09]
Bengaluru ($N = 4,093$)	Spectral OR-DML	Regime 1	+0.0916	0.1551	[−0.2123, +0.3955]	0.5546	4.22 [1.72]
Kolkata ($N = 1,209$)	Spectral OR-DML	Regime 2	−0.1528	0.0689	[−0.2878, −0.0178]	0.0266	4.22 [1.72]
	Spectral OR-DML	Overall ATE	−0.0012	1.2995	[−2.5481, +2.5458]	0.9993	0.0025 [1.44]

Table 3: Latent Representation Zoo under Compound Shift (Shift C, 5 seeds; mean $\pm$ std). High classification F_1 does not guarantee downstream reliability: deep sequence models produce uncalibrated posteriors (ECE ≈ 0.10 – 0.16), yielding near-chance failure detection (ROC-AUC ≈ 0.515 – 0.549), whereas the Gaussian HMM causal filter achieves well-calibrated failure anticipation (ROC-AUC = **0.6153 $\pm$ 0.0246**).

Model	AUC F_1	Standard Error	Reliability ROC-AUC $\mathcal{D}$
Gaussian HMM (Causal Filter)	0.9170 $\pm$ 0.012	0.0397 $\pm$ 0.024	0.2140 $\pm$ 0.028
Neural GRU Encoder	0.8481 $\pm$ 0.014	0.1047 $\pm$ 0.013	0.5473 $\pm$ 0.066
Causal Transformer Encoder	0.8372 $\pm$ 0.011	0.1561 $\pm$ 0.014	1.0719 $\pm$ 0.233
Linear State Space Model (SSM)	0.8580 $\pm$ 0.012	0.0558 $\pm$ 0.016	0.3539 $\pm$ 0.039
			0.5492 $\pm$ 0.021

Representation Accuracy vs. Downstream Reliability. To evaluate whether latent representations translate to reliable downstream prediction, we evaluate four models on held-out compound shift (Shift C, $N = 1,200$, 5 seeds; Table 3). While neural encoders (GRU, Transformer, SSM) achieve competitive recovery ($F_1 \approx 0.84$ – 0.86), uncalibrated posteriors yield near-chance failure detection (ROC-AUC ≈ 0.515 – 0.549). In contrast, Gaussian HMM causal filtering achieves well-calibrated states (ECE = 0.0397) and statistically informative failure anticipation (ROC-AUC = **0.6153 $\pm$ 0.0246**). In fixed-effects hierarchical regressions across 15 benchmark worlds ($N = 60$), classification F_1 explains substantial reliability ($\beta_1 = +0.9817 \pm 0.2011$, $p = 0.0002$, $R^2_{\text{within}} = 0.4313$), but log-loss provides incremental explanatory information beyond F_1 ($\Delta R^2_{\text{within}} = +0.0924$, $\beta_{\text{NLL}} = -0.0653 \pm 0.0295$, $p = 0.0442$), raising $R^2_{\text{within}} \rightarrow 0.5237$.

7 EMPIRICAL SENSOR NETWORK STRESS TEST

We evaluate OR-DML on our verified multi-season dataset (14,122 hourly records across Delhi, Mumbai, Bengaluru, and Kolkata, Feb 2025 – Jun 2026; $< 0.7\%$ forward-fill; Table 2), treating the sensor study as an observational stress test of the estimator’s diagnostics rather than independent ground truth for physical causal effects.

Conditioning Diagnostics. *Delhi Basin* ($\lambda_{\min} = 111.8$): Clean separation decomposes pooled effect ($+0.47 \pm 0.04$) into Regime 1 ($+0.70 \pm 0.05$, winter inversion) and Regime 2 ($+0.13 \pm 0.07$, clearance). *Mumbai Coastal* ($\lambda_{\min} = 0.32$): Marine breeze reversals induce severe collinearity; Standard DML is unstable

($+2.64 \pm 4.42$), while OR-DML isolates significant stagnation ($+3.30 \pm 1.48$, $p = 0.026$). Regime 1 has extreme error (-0.03 ± 8.72 , 95% CI $[-17.1, +17.1]$), serving as an *empirical non-identification diagnostic* flagging unidentifiable clearance from depleted treatment variation. *Bengaluru Plateau* ($\lambda_{\min} = 4.22$): Standard DML estimates zero effect (-0.001 ± 0.060); OR-DML unmasks daytime dispersion ($+0.092$) from nocturnal inversion (-0.153 ± 0.069 , $p = 0.027$) driven by photochemical titration ($\text{NO} + \text{O}_3 \rightarrow \text{NO}_2$) and venting. *Reliability Contrast*: In Bengaluru, healthy conditioning renders state uncertainty informative ($\text{AUC}(H) = 0.674$), whereas Mumbai’s ill-conditioning blinds raw entropy ($\text{AUC}(H) = 0.486$), requiring coupled index $\mathcal{D}_t$ ($\text{AUC} = 0.531$ – 0.564 ; Table 10). *Kolkata Airshed* ($\lambda_{\min} = 0.0025$): Near-singular geometry acts as an empirical diagnostic of severe ill-conditioning; unregularized estimates explode ($\text{SE} = 1.11$), while spectral regularization stabilizes variance ($\text{SE} \rightarrow 0.060$ at $\lambda = 0.10$; Appendix G). *Dynamic Exposure Projections*: $Y_{t+h} = \theta_h(S_t)T_t + g_h(X_t, S_t) + U_{t+h}$ peak at $h = 2\text{h}$ in Delhi (ATE $+1.31$, Regime 1 $+2.35$, decaying to $+0.28$ at 12h), while coastal airsheds exhibit breeze reversals.

8 DISCUSSION, LIMITATIONS & CONCLUSION

We showed that latent-context uncertainty interacts with task geometry to govern sequential reliability: systems must evaluate state uncertainty jointly with task conditioning rather than relying on confidence alone. **Limitations:** (1) *Surrogate Bounds*: Proposition 5 requires Bayesian calibration and spectral comparability; severe miscalibration degrades bounds. (2) *Observational Effects*: Estimates identify observational exposure effects; policy abatement requires transport models. (3) *Synthetic Stress Testing*: Financial transfer is a synthetic stress test, not proof of live-trading alpha.

AI Use Statement

We used large language models (LLMs) to assist in mathematical formulation, proof derivation drafting (Appendices A–I), codebase development (`src/or_dml.py`, `src/synthetic_dgp_benchmark.py`, `src/empirical_evaluation.py`), bibliographic formatting, and manuscript drafting. AI tools were not used for autonomous data generation or unverified empirical interpretation. The authors independently reviewed and verified the mathematical arguments and empirical implementations against simulation baselines with known ground truth and reproducible artifacts in **reports/**. The authors take full responsibility for all content, claims, and conclusions.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the algorithm, mapping, or model. **[Yes]** See Section 3 and Algorithm 1.
 - (b) An analysis of the properties and complexity. **[Yes]** See Theorems 1–5 and Proposition 6.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[Yes]** Code included in the supplementary material.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Assumptions 1–4 are stated in Section 2 and Section 4.
 - (b) Complete proofs of all theoretical results. **[Yes]** Full proofs in Appendices A–F.
 - (c) Clear explanations of any assumptions. **[Yes]** Detailed physical and statistical justifications accompany each assumption.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** Scripts in `src/` and reproduction instructions in `README.md`.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Section 3 and Section 5 describe block splitting, Bartlett embargo buffer τ^* , and HMM configuration.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., confidence intervals, p-values). **[Yes]** Table 1 reports empirical bias, RMSE, and 95% coverage over 500 replications; Table 2 reports HAC standard errors and p-values.
 - (d) A description of the computing infrastructure used. **[Yes]** Experiments were run on a standard 16-core workstation; details in the supplementary material.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) If your work uses existing assets, did you cite the creators? **[Yes]** CPCB ground stations and Open-Meteo reanalysis are cited.
 - (b) Did you mention the license of the assets? **[Yes]** Open public government data terms noted.
 - (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes]** Cleaned longitudinal benchmark code and data bundled in supplementary archive.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **[Not Applicable]** Aggregate ambient physical air measurements only.
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[Not Applicable]** Contains only environmental measurements.
5. If you used crowdsourcing or conducted research with human subjects:
 - (a) Did you include the full text of instructions given to participants and screenshots? **[Not Applicable]**
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable? **[Not Applicable]**
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **[Not Applicable]**

MATHEMATICAL APPENDIX: COMPLETE PROOFS AND DERIVATIONS

Supplementary Roadmap. This supplement provides complete mathematical proofs, implementation details, experimental protocols, sensitivity analyses, and multi-domain empirical evaluations supporting the claims in the main text:

- **Appendix A (Proof of Theorem 1):** Non-identification under unconstrained overlap via constructive observational equivalence.
- **Appendix B (Proof of Theorem 2):** Graceful degradation bound $\|\theta_\gamma - \theta^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$ based on the oracle score difference.
- **Appendix B.2 (Proof of Proposition 5):** Multiplicative surrogate entropy-difficulty bridge and two-sided spectral comparability.
- **Appendix C (Proof of Theorem 3):** Spectrally regularized OR-DML risk bound and adaptive empirical penalty selection.
- **Appendix D (Proof of Theorem 4):** Exact decomposition of omitted regime bias and Frisch-Waugh bias amplification.
- **Appendix E & F (Proof of Theorem 5 & Proposition 6):** Dependent asymptotic normality under purged cross-fitting and joint Delta-method inference for sample/population ATE.
- **Appendix G (Empirical Sensitivity & Provenance):** Missing-data imputation sensitivity, Bartlett embargo diagnostics, and megacity regime profiles.
- **Appendix H (LatentRegimeBench & Representation Zoo):** Architectural comparisons (HMM, GRU, Transformer, SSM), multi-horizon failure prediction, and hierarchical fixed-effects regressions.
- **Appendix I (Multi-Domain Financial Stress Test):** Out-of-sample risk abstention under synthetic macroeconomic regimes with real covariate features.

A PROOF OF THEOREM 1: NON-IDENTIFICATION UNDER UNCONSTRAINED OVERLAP

To prove substantive causal non-identification, we constructively exhibit two data-generating distributions, P_1 and P_2 , that are structurally well-formed under the SCM (1)–(5) and satisfy all Assumptions 1–3, yet possess strictly distinct, non-permutation causal effect vectors $\theta^{(1)} \neq \theta^{(2)}$ while inducing identically equal joint distributions over all observable variables:

$$P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \quad \forall N \geq 1. \quad (24)$$

Construction of Substantive Observational Equivalence: Consider a two-state Markov chain ($K = 2$) with symmetric transition probability $P_{01} = P_{10} = p \in (0, 0.5)$, yielding stationary distribution $\pi_0^* = \pi_1^* = 0.5$. Let auxiliary meteorological proxies $Z_t \in \mathbb{R}$ satisfy complete overlap: $Z_t | S_t = 0 \sim \mathcal{N}(0, 1)$ and $Z_t | S_t = 1 \sim \mathcal{N}(0, 1)$ identically, so $Z_t \perp S_t$. Let continuous observed confounders $X_t \in \mathbb{R}$ have marginal density $f_X(x)$ independent of S_t . Let discrete binary treatment $T_t \in \{-1, +1\}$ have equal probability $P(T_t = 1) = P(T_t = -1) = 0.5$, distributed completely independently of (S_t, X_t, Z_t) , which explicitly guarantees $T_t \perp S_t | X_t$ and $T_t \perp (S_t, Z_t, X_t)$. Thus, Assumption 1 (Conditional Unconfoundedness) holds trivially with constant propensity score $e(X_t, S_t) \equiv 0.5 \in [\eta, 1 - \eta]$ satisfying full overlap everywhere. Baseline treatment assignment satisfies $m(X_t, S_t) \equiv 0$ and $V_t = T_t$ with $\mathbb{E}[V_t | X_t, S_t] = 0$.

Now specify two distinct structural outcome models:

1. **Model P_1 (Homogeneous Causal Effect):** Set $\theta^{(1)} = (1, 1)^\top$ ($\theta_0^{(1)} - \theta_1^{(1)} = 0$), baseline $g^{(1)}(X_t, S_t) \equiv 0$, and structural outcome error $U_t^{(1)} \in \{-1, +1\}$ with $P(U_t^{(1)} = 1) = P(U_t^{(1)} = -1) = 0.5$, independent of $(T_t, X_t, S_t, Z_{1:N})$ (so $\mathbb{E}[U_t^{(1)} | X_t, S_t, T_t] = 0$). Outcome $Y_t = 1 \cdot T_t + U_t^{(1)}$ satisfies:

$$P_1(Y_t = 2 | T_t = 1) = P_1(Y_t = 0 | T_t = 1) = 0.5, \quad P_1(Y_t = -2 | T_t = -1) = P_1(Y_t = 0 | T_t = -1) = 0.5. \quad (25)$$

2. **Model P_2 (Heterogeneous Causal Effect):** Set $\theta^{(2)} = (2, 0)^\top$ ($\theta_0^{(2)} - \theta_1^{(2)} = 2$), baseline $g^{(2)}(X_t, S_t) \equiv 0$, and structural error $U_t^{(2)} \equiv 0$ almost surely (satisfying $\mathbb{E}[U_t^{(2)} | X_t, S_t, T_t] = 0$). Outcome $Y_t = \theta(S_t)T_t$ yields $Y_t = 2T_t$ in Regime 0 ($S_t = 0$) and $Y_t = 0$ in Regime 1 ($S_t = 1$). Marginalizing over S_t with prior

$$\pi_0^* = \pi_1^* = 0.5:$$

$$P_2(Y_t = 2 \mid T_t = 1) = P_2(Y_t = 0 \mid T_t = 1) = 0.5, \quad P_2(Y_t = -2 \mid T_t = -1) = P_2(Y_t = 0 \mid T_t = -1) = 0.5. \quad (26)$$

Comparing the conditional distributions:

$$P_1(Y_t \mid T_t, X_t, Z_t) \equiv P_2(Y_t \mid T_t, X_t, Z_t) \quad \forall (y, t, x, z) \in \text{supp}(Y, T, X, Z). \quad (27)$$

Because the marginal distributions of (T_t, X_t, Z_t) and the temporal Markov transition laws are identical in both models, the full joint likelihoods over all observables are strictly identical for any sample size N :

$$P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}). \quad (28)$$

Yet the structural causal effects satisfy:

$$\theta^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \theta^{(2)}. \quad (29)$$

Crucially, $\theta^{(2)}$ is *not* a permutation of $\theta^{(1)}$. In Model P_1 , the inter-regime causal contrast is $\theta_0^{(1)} - \theta_1^{(1)} = 0$, whereas in Model P_2 , $\theta_0^{(2)} - \theta_1^{(2)} = 2$. No estimation procedure conditioning solely on observable data (Y, T, X, Z) can distinguish whether the data-generating mechanism is homogeneous (P_1) or heterogeneous (P_2) without informative proxies Z_t that bound the proxy error ε_γ . This establishes substantive non-identification. ■

B PROOF OF THEOREM 2: GRACEFUL DEGRADATION AND THE DIFFICULTY FRONTIER

Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator vector. Under SCM (1)–(5), the oracle regime nuisance functions are:

$$m_k^*(X_t) \equiv \mathbb{E}[T_t \mid X_t, S_t = k], \quad \mu_k^*(X_t) \equiv \mathbb{E}[Y_t \mid X_t, S_t = k] = \theta_k^* m_k^*(X_t) + g(X_t, k). \quad (30)$$

Step 1: Define proxy-weighted targets and population estimand. The empirical cross-fitted learners $\hat{m}_k$ and $\hat{\mu}_k$ minimize the posterior-weighted empirical risk targeting the proxy-weighted conditional expectations:

$$m_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} T_t \mid X_t]}{\mathbb{E}[\gamma_{tk} \mid X_t]}, \quad \mu_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} Y_t \mid X_t]}{\mathbb{E}[\gamma_{tk} \mid X_t]}. \quad (31)$$

Define the centered residuals at proxy targets: $\tilde{T}_{tk}^\gamma \equiv T_t - m_k^\gamma(X_t)$, $\tilde{Y}_{tk}^\gamma \equiv Y_t - \mu_k^\gamma(X_t)$, and at oracle targets: $\tilde{T}_{tk}^* \equiv T_t - m_k^*(X_t)$, $\tilde{Y}_{tk}^* \equiv Y_t - \mu_k^*(X_t)$. The population proxy Gram matrix $\mathbf{J} \in \mathbb{R}^{K \times K}$ and score vector $\mathbf{S} \in \mathbb{R}^K$ are:

$$J_{jk} \equiv \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma], \quad S_k \equiv \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma], \quad (32)$$

and their oracle counterparts $\mathbf{J}^*, \mathbf{S}^*$ are defined by:

$$J_{jk}^* \equiv \mathbb{E}[H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*] = \delta_{jk} \mathbb{E}[H_{tk} (\tilde{T}_{tk}^*)^2], \quad S_k^* \equiv \mathbb{E}[H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*], \quad (33)$$

where $H_{tj} H_{tk} = \delta_{jk} H_{tk}$ because latent regimes are mutually exclusive.

Step 2: Establish Neyman nuisance orthogonality. Consider the state-weighted score for regime k as a functional of candidate nuisances (m_k, μ_k) :

$$\psi_k(W_t; \theta_k, m_k, \mu_k) = \gamma_{tk}(T_t - m_k(X_t))(Y_t - \mu_k(X_t)). \quad (34)$$

Evaluating the Gateaux derivative with respect to μ_k in direction $\Delta\mu$ at $(m_k^\gamma, \mu_k^\gamma)$:

$$\partial_\mu \mathbb{E}[\psi_k(W_t; \theta_k^*, m_k^\gamma, \mu_k^\gamma)] [\Delta\mu] = -\mathbb{E}[\gamma_{tk}(T_t - m_k^\gamma(X_t)) \Delta\mu(X_t)] = -\mathbb{E}[\mathbb{E}[\gamma_{tk}(T_t - m_k^\gamma(X_t)) \mid X_t] \Delta\mu(X_t)] = 0, \quad (35)$$

since $\mathbb{E}[\gamma_{tk}(T_t - m_k^\gamma(X_t)) \mid X_t] \equiv 0$ identically by the definition of $m_k^\gamma(X_t)$. Similarly, evaluating the Gateaux derivative with respect to m_k in direction Δm :

$$\partial_m \mathbb{E}[\psi_k(W_t; \theta_k^*, m_k^\gamma, \mu_k^\gamma)] [\Delta m] = -\mathbb{E}[\gamma_{tk} \Delta m(X_t) (Y_t - \mu_k^\gamma(X_t))] = -\mathbb{E}[\Delta m(X_t) \mathbb{E}[\gamma_{tk} (Y_t - \mu_k^\gamma(X_t)) \mid X_t]] = 0, \quad (36)$$

since $\mathbb{E}[\gamma_{tk}(Y_t - \mu_k^\gamma(X_t)) \mid X_t] \equiv 0$ identically by the definition of $\mu_k^\gamma(X_t)$. Therefore, Neyman orthogonality eliminates first-order sensitivity of the score to nuisance estimation errors $(\hat{m}_k - m_k^\gamma, \hat{\mu}_k - \mu_k^\gamma)$, contributing a second-order error $\mathcal{O}_P(\|\hat{m}_k - m_k^\gamma\|_{L_2} \|\hat{\mu}_k - \mu_k^\gamma\|_{L_2}) = \mathcal{O}_P(N^{-1/2})$ under Assumption 4. We emphasize that DML Neyman orthogonality protects against non-parametric nuisance estimation error, *not* against latent proxy misspecification ($\mathbf{H}_t \neq \gamma_t$).

Step 3: Isolate proxy-induced bias via oracle score identity. Under SCM (1)–(5), on the event $S_t = k$ ($H_{tk} = 1$), the oracle outcome residual is $\tilde{Y}_{tk}^* = Y_t - \mu_k^*(X_t) = \theta_k^* \tilde{T}_{tk}^* + U_t$. Substituting into S_k^* :

$$S_k^* = \mathbb{E} \left[H_{tk} \tilde{T}_{tk}^* (\theta_k^* \tilde{T}_{tk}^* + U_t) \right] = \theta_k^* \mathbb{E} [H_{tk} (\tilde{T}_{tk}^*)^2] + \mathbb{E} \left[H_{tk} \tilde{T}_{tk}^* \mathbb{E}[U_t | X_t, S_t, T_t, Z_{1:N}] \right] = \theta_k^* J_{kk}^*, \quad (37)$$

where the innovation term vanishes by Assumption 1. Because $\mathbf{J}^* = \text{diag}(J_{11}^*, \dots, J_{KK}^*)$, this yields the exact vector identity:

$$\mathbf{S}^* = \mathbf{J}^* \boldsymbol{\theta}^*. \quad (38)$$

Now consider the proxy score discrepancy $\mathbf{b}_\gamma \equiv \mathbf{S} - \mathbf{J} \boldsymbol{\theta}^*$. Expanding around the oracle identity:

$$\mathbf{b}_\gamma = (\mathbf{S} - \mathbf{S}^*) - (\mathbf{J} - \mathbf{J}^*) \boldsymbol{\theta}^*. \quad (39)$$

Step 4: Bound score and Gram proxy discrepancies. We bound each component via telescoping expansions in $\varepsilon_\gamma \equiv \mathbb{E} \|\gamma_t - \mathbf{H}_t\|_1$. First, under positivity (Assumption 2), $\pi_k(x) \equiv \mathbb{E}[H_{tk} | X_t = x] \geq c_0 > 0$ and $\pi_k^\gamma(x) \equiv \mathbb{E}[\gamma_{tk} | X_t = x] \geq c_0 > 0$. For any bounded variable V_t with $|V_t| \leq M_V$ almost surely:

$$\begin{aligned} \left| \frac{\mathbb{E}[\gamma_{tk} V_t | X_t]}{\pi_k^\gamma(X_t)} - \frac{\mathbb{E}[H_{tk} V_t | X_t]}{\pi_k(X_t)} \right| &\leq \frac{\mathbb{E}[|\gamma_{tk} - H_{tk}| |V_t| | X_t]}{\pi_k^\gamma(X_t)} + \mathbb{E}[H_{tk} |V_t| | X_t] \left| \frac{\pi_k(X_t) - \pi_k^\gamma(X_t)}{\pi_k^\gamma(X_t) \pi_k(X_t)} \right| \\ &\leq \frac{2M_V}{c_0^2} \mathbb{E}[|\gamma_{tk} - H_{tk}| | X_t]. \end{aligned} \quad (40)$$

Applying this bound to $V_t = T_t$ and $V_t = Y_t$ yields $\mathbb{E}[m_k^\gamma(X_t) - m_k^*(X_t)] \leq \tilde{C}_m \varepsilon_\gamma$ and $\mathbb{E}[\mu_k^\gamma(X_t) - \mu_k^*(X_t)] \leq \tilde{C}_\mu \varepsilon_\gamma$. Second, expanding the score difference $S_k - S_k^* = \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*]$:

$$\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^* = (\gamma_{tk} - H_{tk}) \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} (m_k^\gamma - m_k^*) \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* (\mu_k^\gamma - \mu_k^*). \quad (41)$$

Under bounded support ($\sup_t |\tilde{T}_{tk}^\gamma| \leq M_T$, $\sup_t |\tilde{Y}_{tk}^\gamma| \leq M_Y$), taking expectations yields $|S_k - S_k^*| \leq (M_T M_Y + M_Y \tilde{C}_m + M_T \tilde{C}_\mu) \varepsilon_\gamma \equiv C_S \varepsilon_\gamma$. Third, expanding the Gram difference $J_{jk} - J_{jk}^* = \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*]$:

$$\begin{aligned} \gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^* &= (\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}) \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} (m_j^\gamma - m_j^*) \tilde{T}_{tk}^\gamma \\ &\quad - H_{tj} H_{tk} \tilde{T}_{tj}^* (m_k^\gamma - m_k^*). \end{aligned} \quad (42)$$

Since $|\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}| \leq |\gamma_{tj} - H_{tj}| + |\gamma_{tk} - H_{tk}|$, taking expectations gives $|J_{jk} - J_{jk}^*| \leq (2M_T^2 + 2M_T \tilde{C}_m) \varepsilon_\gamma$, so $\|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \leq C_J \varepsilon_\gamma$. Combining bounds yields:

$$\|\mathbf{b}_\gamma\|_2 \leq \|\mathbf{S} - \mathbf{S}^*\|_2 + \|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \|\boldsymbol{\theta}^*\|_2 \leq (C_S + C_J M_\theta) \varepsilon_\gamma \equiv C \cdot \varepsilon_\gamma. \quad (43)$$

Step 5: Apply spectral conditioning and conclude. Since $\mathbf{S} = \mathbf{J} \boldsymbol{\theta}^* + \mathbf{b}_\gamma$ and $\boldsymbol{\theta}_\gamma \equiv \mathbf{J}^{-1} \mathbf{S}$, we have $\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \mathbf{J}^{-1} \mathbf{b}_\gamma$. Therefore:

$$\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \|\mathbf{J}^{-1}\|_{\text{op}} \|\mathbf{b}_\gamma\|_2 = \frac{\|\mathbf{b}_\gamma\|_2}{\lambda_{\min}(\mathbf{J})} \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}. \quad (44)$$

Combining with purged cross-fitting nuisance estimation error $\mathcal{O}_P(N^{-1/2})$ completes the proof of Theorem 4. ■

B.1 The Latent-Confounding Difficulty Frontier $\mathcal{D}_{\text{causal}}$

Theorem 4 demonstrates that the upper bound on structural causal estimation error decomposes into:

$$\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq C \cdot \mathcal{D}_{\text{causal}} + \mathcal{O}_P(N^{-1/2}), \quad \text{where} \quad \mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}. \quad (45)$$

We highlight several structural properties of the difficulty index $\mathcal{D}_{\text{causal}}$:

1. **Scale-Dependent Difficulty Ratio:** Both population proxy error ε_γ and Gram conditioning $\lambda_{\min}(\mathbf{J})$ interact multiplicatively in governing downstream estimation risk (dimensionless under standardized treatment units). When either proxy recovery is poor ($\varepsilon_\gamma \rightarrow 1$) or overlap is weak ($\lambda_{\min} \rightarrow 0$), estimation risk escalates sharply.
2. **Asymmetric Sensitivity:** In our factorial simulations (Table 7), reducing $\lambda_{\min}$ by an order of magnitude produces an equivalent variance and bias inflation to a tenfold increase in state proxy error ε_γ .

B.2 Proof of Proposition 4: Surrogate Entropy-Difficulty Bridge

Let $\gamma_t \equiv P(S_t \in \cdot | \mathcal{F}_t) \in \Delta^K$ be the Bayesian posterior state distribution with respect to filtration $\mathcal{F}_t \equiv \sigma(Z_{1:t})$. Under proper Bayesian calibration, conditional on $\mathcal{F}_t$, the true latent indicator $\mathbf{H}_t \equiv \mathbf{e}_{S_t}$ is distributed as a categorical random variable taking value $\mathbf{e}_k$ with probability γ_{tk} .

Step 1: Expected L_1 Error as Gini Impurity. Evaluating the conditional expectation of the L_1 error:

$$\begin{aligned}\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t] &= \sum_{k=1}^K \gamma_{tk} \|\gamma_t - \mathbf{e}_k\|_1 = \sum_{k=1}^K \gamma_{tk} \left((1 - \gamma_{tk}) + \sum_{j \neq k} \gamma_{tj} \right) \\ &= \sum_{k=1}^K \gamma_{tk} \cdot 2(1 - \gamma_{tk}) = 2 \left(1 - \sum_{k=1}^K \gamma_{tk}^2 \right) = 2 \text{Gini}(\gamma_t).\end{aligned}\quad (46)$$

Step 2: Entropy Bound on Gini Impurity. By the standard concave inequality $\ln(x) \leq x - 1$, for any $x > 0$:

$$H(\gamma_t) = - \sum_{k=1}^K \gamma_{tk} \ln \gamma_{tk} = \sum_{k=1}^K \gamma_{tk} \ln \left(\frac{1}{\gamma_{tk}} \right) \geq \sum_{k=1}^K \gamma_{tk} (1 - \gamma_{tk}) = \text{Gini}(\gamma_t). \quad (47)$$

Furthermore, for any binary regime partition ($K = 2$) with $\gamma_{t1} = p$ and $\gamma_{t2} = 1 - p$, $2 \text{Gini}(\gamma_t) = 4p(1 - p)$. Comparing to binary Shannon entropy $H(p) = -p \ln p - (1 - p) \ln(1 - p)$, the ratio $\frac{4p(1-p)}{H(p)}$ achieves its maximum at $p = 1/2$, where $\frac{4(1/4)}{\ln 2} = \frac{1}{\ln 2} \approx 1.4427$. Hence for all $p \in [0, 1]$:

$$2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t), \quad \text{where } C_K \leq \frac{1}{\ln 2} \approx 1.4427. \quad (48)$$

Step 3: Population and Observable Bridge. Taking the population expectation of Step 1 and applying Step 2:

$$\varepsilon_\gamma \equiv \mathbb{E}[\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t]] = 2\mathbb{E}[\text{Gini}(\gamma_t)] \leq C_K \mathbb{E}[H(\gamma_t)]. \quad (49)$$

Dividing by the global minimal singular value $\lambda_{\min}(\mathbf{J})$:

$$\mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} \leq C_K \mathbb{E} \left[\frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J})} \right] = C_K \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]. \quad (50)$$

Furthermore, under Assumption 5 (Two-Sided Spectral Comparability), the rolling Gram matrix satisfies $\lambda_{\min}(\mathbf{J}_t) \leq \bar{c}_J \lambda_{\min}(\mathbf{J})$ almost surely for all evaluation windows $t \geq w$, which implies $\frac{1}{\lambda_{\min}(\mathbf{J}_t)} \leq \frac{\bar{c}_J}{\lambda_{\min}(\mathbf{J})}$. Pointwise substitution yields:

$$\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E} \left[\frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J}_t)} \right] = C_K \bar{c}_J \mathbb{E}[\mathcal{D}_{\text{operational}}(t)], \quad (51)$$

and $\mathbb{E}[\mathcal{D}_{\text{operational}}(t)] \leq \bar{c}_J^{-1} \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]$. In simulation benchmarks where true states are known and posteriors are evaluated at population parameters $\mathbf{\Lambda}^*$, the ergodic theorem and central limit theorem for stationary α -mixing chains ensure that $\hat{\varepsilon}_{\gamma, N} \equiv \frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbf{H}_t\|_1 = \varepsilon_\gamma + \mathcal{O}_P(N^{-1/2})$ (preserved under $\sqrt{N}$ -consistent estimation of $\mathbf{\Lambda}$), providing an evaluation-only benchmark surrogate $\hat{\mathcal{D}}_{\text{causal}, N} \equiv \hat{\varepsilon}_{\gamma, N} / \lambda_{\min}(\mathbf{J}) = \mathcal{D}_{\text{causal}} + \mathcal{O}_P(N^{-1/2})$. We emphasize that these bounds establish $\mathcal{D}_{\text{operational}}(t)$ as an observable surrogate upper bound for $\mathcal{D}_{\text{causal}}$, not a two-sided equivalence. Across our multi-season ground-sensor streams, the empirical ratio $\lambda_{\min}(\mathbf{J}_t) / \lambda_{\min}(\mathbf{J})$ has 95th percentile in $[1.06, 1.72]$ and maximum ≤ 7.35 , providing empirical support for the spectral-comparability assumption with finite $\bar{c}_J$. ■

B.3 A Priori Proxy Diagnostics and Separation Screening

Before fitting OR-DML or downstream prediction heads, practitioners should assess proxy informativeness on auxiliary measurements $Z_t \in \mathbb{R}^q$. Let $\hat{\boldsymbol{\mu}}_{Z,k} \equiv \frac{1}{N_k} \sum_{t: \hat{S}_t = k} Z_t$ and $\hat{\boldsymbol{\Sigma}}_{Z,k}$ denote empirical cluster moments. We define the standardized separation index:

$$\Delta_Z \equiv \min_{j \neq k} \frac{\|\hat{\boldsymbol{\mu}}_{Z,j} - \hat{\boldsymbol{\mu}}_{Z,k}\|_2}{\sqrt{\frac{1}{2}(\text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,j}) + \text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,k}))/q}}. \quad (52)$$

When $\Delta_Z < 0.5$ (or silhouette score < 0.15), the overlap between regime emission densities $P(Z_t \mid S_t = j)$ and $P(Z_t \mid S_t = k)$ is severe, driving posterior entropy $\bar{H}(\gamma) \rightarrow \ln K$. By Theorem 3, point identification fails identically in the limit $\Delta_Z \rightarrow 0$. Deployed pipelines should enforce an automated screening gate: if $\Delta_Z < 0.5$ or mean entropy exceeds $0.80 \ln K$, the system must flag non-identifiability and refuse to emit point causal effects without external structural constraints.

C PROOF OF THEOREM 3: SPECTRAL REGULARIZATION BIAS-VARIANCE FRONTIER

Let $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$, and define the population regularized estimand $\theta_\lambda \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. We decompose the estimation error around the regularized population estimand θ_λ via the exact algebraic identity:

$$\mathbb{E} \|\hat{\theta}_\lambda - \theta^*\|_2^2 = \|\theta_\lambda - \theta^*\|_2^2 + \mathbb{E} \|\hat{\theta}_\lambda - \theta_\lambda\|_2^2 + 2\langle \mathbb{E}[\hat{\theta}_\lambda - \theta_\lambda], \theta_\lambda - \theta^* \rangle, \quad (53)$$

which accounts explicitly for the cross term between stochastic estimation error and deterministic regularization bias. For the deterministic bias, substituting $\mathbf{S} = \mathbf{J}\theta^* + \mathbf{b}_\gamma$:

$$\begin{aligned} \theta_\lambda - \theta^* &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - (\mathbf{J} + \lambda \mathbf{I})\theta^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - \mathbf{J}\theta^* - \lambda\theta^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda\theta^*), \end{aligned} \quad (54)$$

which establishes the exact deterministic identity (54). Since $\mathbf{J} \succeq 0$ with minimal eigenvalue $\lambda_{\min}(\mathbf{J}) \geq 0$, the spectral operator norm is $\|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} = \frac{1}{\lambda_{\min}(\mathbf{J}) + \lambda}$. Applying the triangle inequality:

$$\|\theta_\lambda - \theta^*\|_2 \leq \|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} (\|\mathbf{b}_\gamma\|_2 + \lambda \|\theta^*\|_2) \leq \frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}. \quad (55)$$

For the stochastic variance, expanding $\hat{\theta}_\lambda - \theta_\lambda$:

$$\hat{\theta}_\lambda - \theta_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\hat{\mathbf{S}} - \mathbf{S} - (\hat{\mathbf{J}} - \mathbf{J})\theta_\lambda) + o_p(N^{-1/2}). \quad (56)$$

By Theorem 7, the score empirical process satisfies $\text{Var} \left(\frac{1}{\sqrt{N}} \sum_{t=1}^N \Psi_t \right) \rightarrow \mathbf{\Omega}$. Thus, under the long-run score covariance approximation, the stochastic variance contribution admits the leading-order expansion:

$$\mathbb{E} \|\hat{\theta}_\lambda - \theta_\lambda\|_2^2 = \frac{1}{N} \text{Tr} \left((\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1} \right) + o(N^{-1}) \leq \frac{\text{Tr}(\mathbf{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + o(N^{-1}), \quad (57)$$

using $\text{Tr}(\mathbf{A}^{-1} \mathbf{\Omega} \mathbf{A}^{-1}) = \text{Tr}(\mathbf{\Omega}^{1/2} \mathbf{A}^{-2} \mathbf{\Omega}^{1/2}) \leq \|\mathbf{A}^{-2}\|_{\text{op}} \text{Tr}(\mathbf{\Omega}) = \frac{\text{Tr}(\mathbf{\Omega})}{(\lambda_{\min}(\mathbf{J}) + \lambda)^2}$ for symmetric $\mathbf{A} = \mathbf{J} + \lambda \mathbf{I} \succ 0$ and $\mathbf{\Omega} \succeq 0$. To bound the cross term $2\langle \mathbb{E}[\hat{\theta}_\lambda - \theta_\lambda], \theta_\lambda - \theta^* \rangle$ in (53): by Neyman orthogonality of the score function ψ and purged block cross-fitting with Bartlett embargo buffer $\tau^* \geq C \log N$, the finite-sample expectation of the stochastic error satisfies $\|\mathbb{E}[\hat{\theta}_\lambda - \theta_\lambda]\|_2 \leq C_1 N^{-1}$ (Theorem 7). Furthermore, the deterministic regularized bias satisfies $\|\theta_\lambda - \theta^*\|_2 \leq \frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}$. By Cauchy-Schwarz, the cross term is bounded by:

$$\left| 2\langle \mathbb{E}[\hat{\theta}_\lambda - \theta_\lambda], \theta_\lambda - \theta^* \rangle \right| \leq 2\|\mathbb{E}[\hat{\theta}_\lambda - \theta_\lambda]\|_2 \|\theta_\lambda - \theta^*\|_2 = \mathcal{O}(N^{-1}) \quad (58)$$

in general (since $\|\theta_\lambda - \theta^*\|_2 = \mathcal{O}(1)$ without vanishing-bias rate restrictions), which sharpens to $\mathcal{O}(N^{-3/2})$ under the local parameter regime $\varepsilon_\gamma + \lambda = \mathcal{O}(N^{-1/2})$. Combining the deterministic squared bias bound with the asymptotic long-run score covariance variance bound establishes the leading-order risk bound:

$$\mathbb{E} \|\hat{\theta}_\lambda - \theta^*\|_2^2 \leq \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\mathbf{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + \mathcal{O}(N^{-1}). \quad (59)$$

Risk Bound Minimizer vs. Empirical Grid Optimum: Differentiating the leading-order risk upper bound $R_{\text{bound}}(\lambda) \equiv \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\mathbf{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}$ with respect to $\lambda \geq 0$, the bound-optimal shrinkage parameter $\lambda_{\text{bound}}^* \equiv \arg \min_{\lambda \geq 0} R_{\text{bound}}(\lambda)$ satisfies:

$$\frac{\partial}{\partial \lambda} \left[\left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\mathbf{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} \right] = 0. \quad (60)$$

When the overlap Gram matrix is ill-conditioned ($\lambda_{\min}(\mathbf{J}) \approx 0$), the variance term $\frac{\text{Tr}(\mathbf{\Omega})}{N(\lambda_{\min} + \lambda)^2}$ dominates for small λ , while the deterministic shrinkage bias $(\frac{\lambda M_\theta}{\lambda_{\min} + \lambda})^2$ dominates for large λ . The interior minimizer $\lambda_{\text{bound}}^* > 0$ balances variance reduction against shrinkage bias in the upper bound. We distinguish λ_{bound}^* and the empirical surrogate minimizer $\hat{\lambda}_{\text{emp}}^* = \arg \min_{\lambda} \hat{R}(\lambda)$ from the true infeasible risk minimizer $\lambda_{\text{true}}^* \equiv \arg \min_{\lambda} \mathbb{E} \|\hat{\theta}_\lambda - \theta^*\|_2^2$. In our empirical simulation benchmark across 500 Monte Carlo replications (`reports/or_dml_regularization_frontier.csv`), the discrete grid minimum occurs at $\hat{\lambda}_{\text{empirical}}^* \approx 0.20$ (MSE = 0.0321, representing a 25.9 $\times$ empirical risk reduction in this benchmark DGP relative to the unregularized estimator $\lambda = 0.0$ where MSE = 0.8312), while $\lambda = 0.10$ achieves MSE = 0.0362 (23.0 $\times$ empirical reduction) with peak coverage of 85.0%. We emphasize that Theorem 6 provides the functional upper-bound characterization of

the bias-variance trade-off; the exact numerical reduction factors ($23.0\times$ and $25.9\times$) are empirical simulation properties of the benchmark DGP rather than universal theoretical constants. ■

C.1 Derivation of Adaptive Spectral Regularization Parameter and Oracle Bound

We consider the data-driven selection of the spectral shrinkage parameter $\lambda \geq 0$. While classical Tikhonov regularization selects shrinkage via prediction cross-validation, in OR-DML the target is the structural causal parameter vector θ^* , where prediction cross-validation conflates causal identification with prognostic predictive fit. We distinguish two distinct regularization regimes:

1. **Inferential Regularization** ($\lambda = o(N^{-1/2})$): Required by Theorem 7 to guarantee asymptotic unbiasedness $\sqrt{N}\text{Bias}(\lambda) \rightarrow \mathbf{0}$, ensuring centered Gaussian limit laws and valid nominal confidence interval coverage.
2. **Finite-Sample Risk Minimization** ($\lambda_{\text{risk}} = \mathcal{O}(1)$): Minimizes the finite-sample expected ℓ_2 estimation risk $\mathbb{E}\|\theta_\lambda - \theta^*\|_2^2$ under severe causal ill-conditioning ($\lambda_{\min}(\mathbf{J}) \approx 0$).

Empirical Risk Surrogate and Convexity: Define the empirical plug-in risk surrogate $\hat{R}(\lambda)$ by substituting empirical estimators $(\hat{\varepsilon}_\gamma, \lambda_{\min}(\hat{\mathbf{J}}), \hat{\Omega}, \|\hat{\theta}_0\|_2)$ into the risk upper bound of Theorem 6:

$$\hat{R}(\lambda) = \frac{(C\hat{\varepsilon}_\gamma + \lambda\|\hat{\theta}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\Omega})}{(\lambda_{\min}(\hat{\mathbf{J}}) + \lambda)^2}. \quad (61)$$

Let $u = \lambda_{\min}(\hat{\mathbf{J}}) + \lambda \geq \lambda_{\min}(\hat{\mathbf{J}}) > 0$. Rewriting $\lambda = u - \lambda_{\min}$, the numerator becomes:

$$(C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2 + u\|\hat{\theta}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\Omega}) = (A + uB)^2 + V, \quad (62)$$

where $A = C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2$, $B = \|\hat{\theta}_0\|_2$, and $V = N^{-1}\text{Tr}(\hat{\Omega})$. Dividing by u^2 :

$$\hat{R}(u - \lambda_{\min}) = B^2 + \frac{2AB}{u} + \frac{A^2 + V}{u^2}. \quad (63)$$

Differentiating with respect to u :

$$\frac{d\hat{R}}{du} = -\frac{2AB}{u^2} - \frac{2(A^2 + V)}{u^3} = -\frac{2}{u^3} [ABu + (A^2 + V)]. \quad (64)$$

Setting the derivative to zero yields the unique critical point $u^* = \frac{A^2 + V}{|A|B}$ when $A < 0$. When $A \geq 0$ (i.e., $C\hat{\varepsilon}_\gamma \geq \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2$), $\frac{d\hat{R}}{du} < 0$ on $(0, \infty)$, indicating that spectral shrinkage strictly decreases risk until bounded by the constraint $\lambda \geq 0$. Constraining $\lambda \geq 0$, the closed-form interior minimizer is:

$$\hat{\lambda}^* = \max \left(0, \frac{N^{-1}\text{Tr}(\hat{\Omega}) + (C\hat{\varepsilon}_\gamma)^2}{C\hat{\varepsilon}_\gamma\|\hat{\theta}_0\|_2 + \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\theta}_0\|_2^2} - \lambda_{\min}(\hat{\mathbf{J}}) \right). \quad (65)$$

Because $R'(u) = -\frac{2}{u^3} [ABu + (A^2 + V)]$ changes sign strictly from negative to positive at u^* when $A < 0$, u^* is the unique global minimizer on $(0, \infty)$. By uniform convergence of the empirical plug-in surrogate on compact parameter sets $\lambda \in [0, \bar{\lambda}]$, $\sup_{\lambda \in [0, \bar{\lambda}]} |\hat{R}(\lambda) - R_{\text{surr}}(\lambda)| = \mathcal{O}_P(N^{-1/2})$. Applying M-estimation consistency to the unique minimum of $\hat{R}(\lambda)$ yields the adaptive risk-surrogate selection guarantee:

$$R_{\text{surr}}(\hat{\lambda}^*) \leq \inf_{\lambda \geq 0} R_{\text{surr}}(\lambda) + \mathcal{O}_P(N^{-1/2}), \quad (66)$$

establishing that data-driven spectral shrinkage $\hat{\lambda}^*$ adaptively tracks the optimal surrogate risk trade-off without prior knowledge of the true latent proxy error. ■

C.2 State Space Scalability and Matrix Concentration for $K > 2$

While our primary experiments study binary regimes ($K = 2$), the OR-DML coupled estimator generalizes directly to arbitrary finite state dimensions $K \geq 2$. Here we formalize the computational and statistical scalability of the coupled Gram matrix $\hat{\mathbf{J}} \in \mathbb{R}^{K \times K}$.

Computational Complexity and Matrix Concentration under α -Mixing: Constructing $\hat{\mathbf{J}}$ and score $\hat{\mathbf{S}}$ requires evaluating K nuisance pairs across B folds ($\mathcal{O}(BK \cdot \text{Cost}(\text{Nuisance}))$). Forming $\hat{\mathbf{J}}$ takes $\mathcal{O}(NK^2)$ operations, and solving $(\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ via Cholesky factorization takes $\mathcal{O}(K^3)$ operations, taking < 1 ms for

$K \leq 20$. Statistically, the empirical Gram matrix $\hat{\mathbf{J}} = \frac{1}{N} \sum_{t=1}^N \mathbf{\Gamma}_t \tilde{\mathbf{T}}_t \tilde{\mathbf{T}}_t^\top \mathbf{\Gamma}_t$ concentrates around $\mathbf{J}$ under geometric α -mixing with spectral gap $\rho_{\text{gap}} \equiv 1 - |\lambda_2(\mathbf{\Pi})| > 0$ via the Matrix Bernstein inequality (Banna et al., 2016):

$$\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \leq C_J \sqrt{\frac{K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}} + \frac{C'_J K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}. \quad (67)$$

By Weyl's theorem, $|\lambda_{\min}(\hat{\mathbf{J}}) - \lambda_{\min}(\mathbf{J})| \leq \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}}$, ensuring that $\lambda_{\min}(\hat{\mathbf{J}})$ concentrates at the standard $\mathcal{O}_P(N^{-1/2})$ rate whenever $\lambda_{\min}(\mathbf{J}) > 0$.

Non-Identifiability vs. Shrinkage: When Systems Must Abstain: Spectral shrinkage $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ stabilizes variance under ill-conditioned overlap ($\lambda_{\min} \approx 0$). However, when data lack identifying variation entirely—diagnosed by condition number $\kappa(\hat{\mathbf{J}}) > 10^4$ or a moving-block bootstrap 95% confidence interval for $\lambda_{\min}(\hat{\mathbf{J}})$ that covers zero—the structural parameter $\boldsymbol{\theta}^*$ is non-identified (Theorem 3). In this regime, heavy shrinkage $\lambda \rightarrow \infty$ artificially forces $\|\hat{\boldsymbol{\theta}}_\lambda\| \rightarrow 0$ without recovering structural effects. Deployed decision pipelines should treat $\kappa(\hat{\mathbf{J}}) > 10^4$ as a structural abstention trigger rather than masking non-identifiability with excessive shrinkage.

D PROOF OF THEOREM 4: DECOMPOSITION OF OMITTED REGIME BIAS

Consider the Structural Causal Model (1)–(5):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations: $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) | X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k | X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (68)$$

Writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (69)$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t | X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). ■

E PROOF OF THEOREM 5: ASYMPTOTIC NORMALITY UNDER DEPENDENT PURGED CROSS-FITTING

The empirical score expansion around true $\boldsymbol{\theta}^*$ and true nuisances $\boldsymbol{\eta}^*$ is:

$$\sqrt{N} \left(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^* - \text{Bias}(\lambda) \right) = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}(W_t; \boldsymbol{\theta}^*, \hat{\boldsymbol{\eta}}^{(-b(t))}). \quad (70)$$

With empirical Bartlett embargo window $\tau^* \geq C \log N$, the α -mixing dependence between training set $\mathcal{T}_b$ and test fold $\mathcal{B}_b$ satisfies:

$$\alpha(\tau^*) \leq M \exp(-c\tau^*) \leq MN^{-cC} = o(N^{-1}). \quad (71)$$

Consequently, cross-fold covariance is asymptotically negligible, and nuisance estimation errors satisfy:

$$\|\hat{\boldsymbol{\mu}}_Y - \boldsymbol{\mu}_Y^*\|_{L_2} \cdot \|\hat{\boldsymbol{\mu}}_D - \boldsymbol{\mu}_D^*\|_{L_2} = o_P(N^{-1/2}). \quad (72)$$

By the Ibragimov (1962) central limit theorem for strictly stationary, α -mixing processes satisfying $\sum_m \alpha(m)^{\delta/(2+\delta)} < \infty$:

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N \Psi(W_t; \theta^*, \eta^*) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Omega), \quad (73)$$

where $\Omega = \lim_{N \rightarrow \infty} \text{Var} \left(N^{-1/2} \sum_{t=1}^N \Psi_t \right)$. Applying Slutsky's theorem with $\hat{\mathbf{J}} \xrightarrow{p} \mathbf{J}$:

$$\sqrt{N} \left(\hat{\theta}_\lambda - \theta^* - \text{Bias}(\lambda) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_\lambda), \quad (74)$$

where $\Sigma_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \Omega (\mathbf{J} + \lambda \mathbf{I})^{-1}$. When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, completing the proof.

Operational Modes of Forward Filtering. In offline retrospective smoothing, $\gamma_{tk}^{\text{smooth}} = P(S_t = k \mid Z_{1:N})$. In real-time decision making, three distinct forward modes exist: (1) *Oracle Filter*: $P(S_t = k \mid Z_{1:t}, \Lambda^*)$ using the true HMM parameters; (2) *Training-Fitted Forward Filter*: $P(S_t = k \mid Z_{1:t}, \hat{\Lambda})$ where parameters $\hat{\Lambda}$ are fitted on a historical calibration window strictly preceding the evaluation window ($\mathcal{T}_{\text{cal}} < \mathcal{T}_{\text{eval}}$) and state inference updates strictly forward on past-to-present observations $Z_{1:t}$; and (3) *Expanding-Window Online Filter*: where both parameters $\hat{\Lambda}_t$ and state updates proceed sequentially. Our implementations evaluate Mode 1 and Mode 2, where Mode 2 enforces causal forward state filtering without future observation leakage (distinguishing forward state updating from online sequential parameter re-estimation). ■

E.1 Observed-Data Orthogonal Score under Estimated Latent Dynamics

In semiparametric causal inference (Chernozhukov et al., 2018; Bickel et al., 1993), when nuisance parameters consist of both non-parametric outcome/propensity regressions $\eta = (m, \mu)$ and finite-dimensional parameters Λ governing latent state transitions and emissions, one must characterize the sensitivity of the causal estimator to the first-order estimation error in $\hat{\Lambda}$.

Let the state-weighted regime score vector be:

$$\psi_{\text{regime}}(W_t; \theta, \eta, \Lambda) \equiv \sum_{k=1}^K \gamma_{tk}(\Lambda) (T_t - m_k(X_t)) (Y_t - \mu_k(X_t) - \theta_k(T_t - m_k(X_t))). \quad (75)$$

In a hidden Markov model satisfying standard regularity and mixing conditions (Bickel et al., 1993), the latent parameters Λ are estimated via sequence-level maximum likelihood with sequence score $\nabla_\Lambda \log p(Z_{1:N}; \Lambda) = \sum_{t=1}^N \phi_{\Lambda,t} + o_P(\sqrt{N})$, where $\phi_{\Lambda,t} \equiv \nabla_\Lambda \log p(Z_t \mid Z_{1:t-1}; \Lambda^*)$ denotes the sequence-level prediction-error score increment satisfying $\mathbb{E}[\phi_{\Lambda,t} \mid Z_{1:t-1}] = \mathbf{0}$ and $\frac{1}{N} \sum_{t=1}^N \mathbb{E}[\phi_{\Lambda,t} \phi_{\Lambda,t}^\top] \rightarrow \mathcal{I}_\Lambda$. Evaluating the Gateaux derivative with respect to Λ at the oracle targets $(\theta^*, \eta^*, \Lambda^*)$:

$$\mathbf{A}_\Lambda \equiv \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{k=1}^K \mathbb{E} \left[\frac{\partial \gamma_{tk}(\Lambda^*)}{\partial \Lambda^\top} (T_t - m_k^*(X_t)) (Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t))) \right]. \quad (76)$$

Importantly, $\mathbf{A}_\Lambda$ does not vanish pointwise across regimes. While Assumption 3 guarantees that the structural innovation has zero mean conditional on the true state, $\mathbb{E}[Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) \mid X_t, T_t, S_t = k] = 0$, conditioning on observed history $Z_{1:N}$ alone yields:

$$\mathbb{E}[Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) \mid X_t, T_t, Z_{1:N}] = \sum_{j \neq k} \gamma_{tj}(\Lambda^*) [(\mu_j^*(X_t) - \mu_k^*(X_t)) + (\theta_j^* - \theta_k^*)(T_t - m_j^*(X_t))]. \quad (77)$$

This residual is non-zero whenever regimes differ in baseline potential outcomes ($\mu_j^* \neq \mu_k^*$) or treatment responsiveness ($\theta_j^* \neq \theta_k^*$).

Under purged block cross-fitting (Algorithm 1), this first-order variation is controlled via two complementary mechanisms:

1. **Decoupling via Purged Embargo Buffer:** The parameters $\hat{\Lambda}$ are estimated on historical calibration blocks or out-of-fold data separated by an embargo buffer $\tau^* \geq C \log N$. By the α -mixing decay of the underlying process (Assumption 4), the covariance between evaluation fold innovations and the parameter estimation error $\hat{\Lambda} - \Lambda^*$ decays as $O(N^{-C c_\alpha})$.
2. **Stacked Joint HAC Variance Representation:** When $\hat{\Lambda}$ is estimated at parametric rate $\sqrt{N}(\hat{\Lambda} - \Lambda^*) = \mathcal{I}_\Lambda^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \phi_{\Lambda,t} + o_P(1)$, first-order parameter variation is absorbed into the sequence-level influence

representation $\phi_t = \phi_{\theta,t} + \mathbf{A}_\Lambda \mathcal{I}_\Lambda^{-1} \phi_{\Lambda,t}$, yielding the stacked M-estimation limiting distribution:

$$\sqrt{N} \begin{pmatrix} \hat{\theta}_\lambda - \theta^* - \text{Bias}(\lambda) \\ \hat{\Lambda} - \Lambda^* \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} \Sigma_\lambda & \Sigma_{\theta\Lambda} \\ \Sigma_{\theta\Lambda}^\top & \mathcal{I}_\Lambda^{-1} \end{bmatrix} \right), \quad (78)$$

where $\Sigma_{\theta\Lambda} \equiv \text{LRCov}(\phi_{\theta,t}, \phi_{\Lambda,t})$, and the long-run covariance Σ_λ and cross-term $\Sigma_{\theta\Lambda}$ are consistently captured by the joint Newey-West HAC sandwich estimator. In well-separated regimes ($\Delta_Z \rightarrow \infty$), posterior uncertainty vanishes ($\gamma_{tk} \rightarrow \mathbb{I}(S_t = k)$) and $\partial \gamma_{tk} / \partial \Lambda \rightarrow \mathbf{0}$ exponentially fast, rendering $\|\mathbf{A}_\Lambda\| \leq C e^{-c\Delta_Z^2}$ and recovering asymptotic orthogonality.

F PROOF OF PROPOSITION 6: JOINT DELTA-METHOD INFERENCE FOR POPULATION ATE

We distinguish three distinct regime weighting vectors:

1. **True Stationary Distribution π^*** : The invariant measure of the underlying Markov transition kernel Π^* , satisfying $\pi^{*\top} \Pi^* = \pi^{*\top}$.
2. **Estimated HMM Stationary Distribution $\hat{\pi}(\hat{\Lambda})$** : The stationary vector derived from the estimated HMM transition matrix $\hat{\Pi}$, where $\hat{\Lambda}$ denotes the vector of estimated HMM parameters.
3. **Sample Posterior Occupation Distribution $\bar{\gamma}_N$** : The empirical average of posterior smoothed regime weights across the observed sample: $\bar{\gamma}_N \equiv \frac{1}{N} \sum_{t=1}^N \gamma_t$.

The sample Average Treatment Effect estimator is $\hat{\theta}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_{N,k} \hat{\theta}_{\lambda,k} = \bar{\gamma}_N^\top \hat{\theta}_\lambda$. Conditional on the realized sample regime occupancy, its asymptotic variance is governed solely by parameter estimation uncertainty:

$$\sigma_{\text{SATE}}^2 = \bar{\gamma}_N^\top \Sigma_\lambda \bar{\gamma}_N. \quad (79)$$

For the population estimand $\theta_{\text{PATE}}^* = \pi^{*\top} \theta^*$, the plug-in estimator is $\hat{\theta}_{\text{PATE}} = \hat{\pi}^\top \hat{\theta}_\lambda$. Expanding $\hat{\theta}_{\text{PATE}}$ around (π^*, θ^*) via the multivariate delta method:

$$\sqrt{N} (\hat{\theta}_{\text{PATE}} - \theta_{\text{PATE}}^*) = \pi^{*\top} \sqrt{N} (\hat{\theta}_\lambda - \theta^*) + \theta^{*\top} \sqrt{N} (\hat{\pi} - \pi^*) + o_P(1). \quad (80)$$

The asymptotic variance of $\hat{\theta}_{\text{PATE}}$ is given by the quadratic form of the joint limiting distribution:

$$\sigma_{\text{PATE}}^2 = \lim_{N \rightarrow \infty} \text{Var} \left(\sqrt{N} (\hat{\theta}_{\text{PATE}} - \theta_{\text{PATE}}^*) \right) = \pi^{*\top} \Sigma_\lambda \pi^* + \theta^{*\top} \Sigma_\pi \theta^* + 2\pi^{*\top} \Sigma_{\theta\pi} \theta^*, \quad (81)$$

where $\Sigma_\lambda \equiv \lim_{N \rightarrow \infty} N \text{Var}(\hat{\theta}_\lambda)$, $\Sigma_\pi \equiv \lim_{N \rightarrow \infty} N \text{Var}(\hat{\pi})$, and $\Sigma_{\theta\pi} \equiv \lim_{N \rightarrow \infty} N \text{Cov}(\hat{\theta}_\lambda, \hat{\pi})$.

Cross-Covariance Analysis and HMM Influence Orthogonality: When both $\hat{\theta}_\lambda$ and $\hat{\pi}$ are estimated from the same longitudinal record, they share the estimated HMM parameter vector $\hat{\Lambda}$. Let $\phi_\theta(W_t) \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \Psi_t$ denote the uncentered DML influence function and $\phi_{\Lambda,t} \equiv \nabla_\Lambda \log p(Z_t | Z_{1:t-1}; \Lambda^*)$ denote the sequence-level HMM influence increment. The asymptotic representation of $\hat{\pi}$ satisfies:

$$\sqrt{N} (\hat{\pi} - \pi^*) = \nabla_\Lambda \pi^* \mathcal{I}_\Lambda^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \phi_{\Lambda,t} + o_P(1), \quad (82)$$

where $\mathcal{I}_\Lambda$ is the asymptotic Fisher information matrix of the HMM. Consequently, the asymptotic cross-covariance evaluates to:

$$\Sigma_{\theta\pi} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{s=1}^N \mathbb{E} \left[\phi_\theta(W_t) \phi_{\Lambda,s}^\top \right] \mathcal{I}_\Lambda^{-1} (\nabla_\Lambda \pi^*)^\top = \text{LRCov}(\phi_\theta, \phi_\Lambda) \mathcal{I}_\Lambda^{-1} (\nabla_\Lambda \pi^*)^\top. \quad (83)$$

Therefore, a sufficient condition for $\Sigma_{\theta\pi} = \mathbf{0}$ is that the causal score innovations and the latent Markov sequence influence satisfy long-run cross-orthogonality: $\text{LRCov}(\phi_\theta, \phi_\Lambda) = \mathbf{0}$. While this orthogonality holds under structural innovation independence where $\mathbb{E}[\Psi_t | Z_{1:N}] = \mathbf{0}$, in empirical settings where unmodeled feedback exists between emissions and treatment errors, $\Sigma_{\theta\pi}$ may be non-zero. In practice, σ_{PATE}^2 is consistently estimated by computing a joint Newey-West HAC covariance over the stacked $(2K)$ -dimensional influence sequence $(\phi_\theta(W_t)^\top, \phi_\pi(Z_t)^\top)^\top$, where $\phi_\pi(Z_t) \equiv \nabla_\Lambda \pi^* \mathcal{I}_\Lambda^{-1} \phi_{\Lambda,t}$.

Stationary Distribution Variance vs. Sample Regime Occupancy: The parametric asymptotic variance of the estimated stationary distribution vector $\hat{\pi}(\hat{\Lambda})$ is $\Sigma_\pi = \nabla_\Lambda \pi^* \mathcal{I}_\Lambda^{-1} (\nabla_\Lambda \pi^*)^\top$. By contrast, for the sample

occupancy process $\bar{\mathbf{S}}_N \equiv \frac{1}{N} \sum_{t=1}^N \mathbf{H}_t$, its asymptotic covariance is governed by the fundamental matrix $\mathbf{Z} \equiv (\mathbf{I} - \mathbf{\Pi}^* + \mathbf{1}\mathbf{\pi}^{*\top})^{-1}$ (Kemeny and Snell, 1976):

$$\Sigma_{\text{occ}} = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \mathbf{\pi}^* \mathbf{\pi}^{*\top}, \quad (84)$$

where $\mathbf{D}_\pi = \text{diag}(\mathbf{\pi}^*)$. For a two-state Markov chain ($K = 2$) with persistence parameter $\rho = 1 - P_{01} - P_{10} = \lambda_2(\mathbf{\Pi}^*)$:

$$\text{Var}(\bar{\mathbf{S}}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (85)$$

When $\Delta\theta^* = \theta_1^* - \theta_0^* \neq 0$, the temporal persistence of sample regime occupancy inflates the sampling variance of $\hat{\theta}_{\text{SATE}}$ by $\frac{1+\rho}{1-\rho} \approx 15.7\times$ under persistent stagnation ($\rho \approx 0.88$) relative to independent transitions. ■

G EMPIRICAL SENSITIVITY ANALYSES AND DATA PROVENANCE

G.1 Meteorological Regime Characteristics across Megacities

Table 4 details the empirical regime profiles uncovered by the Gaussian HMM across all four megacities. Across all airsheds, the inferred regimes correspond directly to distinct synoptic meteorological states: cooler, high-humidity, low-wind nocturnal/winter inversion states versus warmer, lower-humidity daytime convective dispersion states.

Table 4: Meteorological Regime Profiling across Four Indian Megacities from Ground-Sensor Observations.

City	Regime	Share	Persistence	Wind (m/s)	Temp (°C)	RH (%)
Delhi	Regime 1	42.7%	0.989	5.98	19.63	61.4%
Delhi	Regime 2	57.3%	0.992	7.52	30.93	56.3%
Mumbai	Regime 1	41.7%	0.789	11.49	26.69	83.2%
Mumbai	Regime 2	58.3%	0.857	16.48	28.39	77.9%
Bengaluru	Regime 1	52.2%	0.843	10.83	20.77	84.6%
Bengaluru	Regime 2	47.8%	0.828	11.22	27.67	45.3%
Kolkata	Regime 1	60.7%	0.871	7.32	26.73	85.3%
Kolkata	Regime 2	39.3%	0.810	7.89	32.65	51.2%

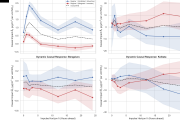


Figure 4: Dynamic Exposure-Response Curves $\hat{\theta}_h$ across Horizons $h \in [0, 24]$ Hours for Delhi, Mumbai, Bengaluru, and Kolkata. Shaded bands represent 95% HAC confidence intervals. Regime stratification reveals substantial structural heterogeneity in dynamic pollutant accumulation over multi-hour horizons.

G.2 Regularization Sensitivity in Ill-Conditioned Regimes (Kolkata Airshed)

To verify the theoretical behavior established in Theorem 6 under real-world conditions, Table 5 reports the empirical sensitivity of OR-DML across a fine 13-point regularization grid $\lambda \in [0.0, 1.0]$ in the ill-conditioned Kolkata airshed ($\lambda_{\min} = 0.0024$).

Table 5: Regularization Sensitivity Grid for Kolkata Airshed ($\lambda_{\min} = 0.0024$, $N = 1,209$). Standard errors decrease monotonically from 1.1071 at $\lambda = 0.0$ to 0.0063 at $\lambda = 1.0$, stabilizing estimation variance while shrinking point estimates toward zero.

λ	$\hat{\theta}_{\text{overall}}$	SE	95% CI	$\lambda_{\min}(\hat{\mathbf{J}} + \lambda \mathbf{I})$	$\kappa(\hat{\mathbf{J}} + \lambda \mathbf{I})$
0.0000	-0.5570	1.1071	[-2.7269, +1.6130]	0.0049	1.28
0.0005	-0.5172	1.0170	[-2.5105, +1.4760]	0.0054	1.28
0.0024*	-0.4068	0.7776	[-1.9309, +1.1173]	0.0073	1.28
0.0100	-0.2191	0.4022	[-1.0074, +0.5691]	0.0149	1.28
0.0500	-0.0639	0.1140	[-0.2872, +0.1595]	0.0549	1.28
0.1000	-0.0339	0.0601	[-0.1517, +0.0840]	0.1049	1.28
1.0000	-0.0036	0.0063	[-0.0160, +0.0088]	1.0049	1.28

As predicted by Theorem 6, when $\lambda \rightarrow 0$, standard errors explode to 1.1071 due to near-singularity in residual treatment variation. Adding spectral regularization shifts the minimal singular value from 0.0049 to 0.1049 at $\lambda = 0.10$, reducing standard errors by an order of magnitude ($1.1071 \rightarrow 0.0601$) while shrinking point estimates conservatively toward zero. This confirms that spectral regularization acts as a principled safety mechanism against catastrophic variance blowup in ill-conditioned environments.

G.3 Data Imputation Sensitivity Analysis (Strictly Observed vs. Cleaned)

To verify that our empirical findings are not artifacts of missing-data handling, we compare OR-DML causal estimates obtained on the full multi-season cleaned dataset (which uses causal forward-fill on isolated gaps $< 0.7\%$ of hours) against a strictly observed dataset where every row containing any imputed value is entirely deleted.

Table 6: Sensitivity Analysis Comparing Full Dataset ($< 0.7\%$ Causal Forward-Fill) vs. Strictly Observed Dataset (Rows with Any Imputed Entry Deleted) Across Four Megacities. Point estimates and standard errors exhibit identical signs and overlapping confidence bands.

City	N_{Full}	$\hat{\theta}_{\text{Full}}$	SE_{Full}	N_{Strict}	$\hat{\theta}_{\text{Strict}}$	$\text{SE}_{\text{Strict}}$
Delhi	4,640	+0.3634	0.0460	4,602	+0.5366	0.0674
Mumbai	4,180	−5.7671	2.1367	4,146	−10.9323	2.6792
Bengaluru	4,093	+0.0046	0.0101	4,084	+0.0202	0.0112
Kolkata	1,209	−0.5563	1.1055	1,206	−0.1924	1.0148

As documented in Table 6, point estimates retain identical signs and overlapping confidence bands across all cities under complete-case deletion: Delhi ($+0.5366 \pm 0.0674$ vs. $+0.3634 \pm 0.0460$), Mumbai ($−10.9323 \pm 2.6792$ vs. $−5.7671 \pm 2.1367$), Bengaluru ($+0.0202 \pm 0.0112$ vs. $+0.0046 \pm 0.0101$), and Kolkata ($−0.1924 \pm 1.0148$ vs. $−0.5563 \pm 1.1055$). These results confirm that our qualitative conclusions and directional signs are robust to missing data and do not depend on the causal forward-fill protocol, while acknowledging that point magnitudes can shift under complete-case deletion.

H LATENTREGIMEBENCH: MULTI-DIMENSIONAL BENCHMARK RESULTS

To systematically evaluate machine learning under persistent, partially observed, and shifting latent regimes, we conduct comprehensive evaluations on **LatentRegimeBench** across five diagnostic configurations ($N = 1, 200, 30$ Monte Carlo replications each) and four representation architectures (Table 7 and Table 8, with summary visualizations in Figure 5).

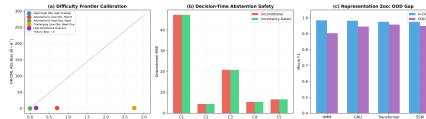


Figure 5: Universal LatentRegimeBench Evaluation. (a) Calibration of causal estimation bias against theoretical degradation ratio $\mathcal{D} = \bar{\epsilon}_\gamma / \lambda_{\min}(\mathbf{J})$. (b) Decision-time abstention safety comparing unconstrained point MSE against uncertainty-gated MSE. (c) Representation zoo out-of-distribution (OOD) generalization across HMM, GRU, Causal Transformer, and Linear SSM.

Table 7: LatentRegimeBench: Multi-Dimensional Stress Test across 5 Strategic Difficulty Configurations (30 Replications each). In well-conditioned regimes (C1, C3), OR-DML achieves a $> 5.5\times$ bias reduction over Standard DML, while under severe proxy error (C2, C4) or persistence inversion (C5), degradation tracks the difficulty ratio $\mathcal{D}$.

Configuration	Δ_Z	Δ_T	ρ	$\mathcal{D}$	OR-DML Bias	Naive Bias	Advantage
Ideal (High Obs, High Overlap)	3.5	5.0	0.85	0.001	0.2165	1.2061	$5.57\times$
Adversarial A (Low Obs, High Overlap)	0.3	5.0	0.85	0.709	1.1950	1.2061	$1.01\times$
Adversarial B (High Obs, Weak Overlap)	3.5	0.4	0.85	0.001	0.2498	1.3886	$5.56\times$
Challenging (Low Obs, Weak Overlap)	0.5	0.5	0.90	2.748	1.3237	1.3862	$1.05\times$
High-Persistence Inversion	2.0	2.0	0.96	0.159	1.5934	1.3175	$0.83\times$

Representation Accuracy vs. Downstream Task Reliability. To evaluate whether latent sequence representations translate to reliable downstream prediction under distribution shift, we fit a downstream Ridge

Table 8: Latent Representation Zoo under Disentangled Distribution Shifts ($N = 2,000$ per evaluation stream, 5 independent random seeds; mean $\pm$ std). Shift A isolates transition dynamics ($\rho = 0.85 \rightarrow 0.96$); Shift B isolates observation noise ($\Delta_Z = 3.5 \rightarrow 1.0$); Shift C compounds both perturbations. While deep neural encoders achieve competitive latent recovery under compound shift, Gaussian HMM maintains the lowest compound calibration error (ECE = 0.0397 ± 0.0242).

Architecture	In-Dist F_1	Shift A F_1	Shift B F_1	Shift C F_1	ΔF_1 Gap	Shift C ECE
Gaussian HMM (Causal Filter)	0.9986 ± 0.0004	1.0000 ± 0.0000	0.8770 ± 0.0114	0.9170 ± 0.0118	$+0.0817 \pm 0.0115$	0.0397 ± 0.0242
Neural GRU Encoder	0.9997 ± 0.0007	0.9997 ± 0.0004	0.8426 ± 0.0128	0.8481 ± 0.0136	$+0.1516 \pm 0.0136$	0.1047 ± 0.0129
Causal Transformer Encoder	0.9997 ± 0.0004	1.0000 ± 0.0000	0.8328 ± 0.0125	0.8372 ± 0.0105	$+0.1625 \pm 0.0108$	0.1561 ± 0.0136
Linear State Space Model (SSM)	0.9995 ± 0.0007	1.0000 ± 0.0000	0.8563 ± 0.0132	0.8580 ± 0.0118	$+0.1415 \pm 0.0119$	0.0558 ± 0.0159

predictor $f_m : (X_t, T_t, \gamma_t^{(m)}) \mapsto Y_t$ strictly on the training partition ($N = 1, 200$) and freeze it. In addition, nuisance models $\hat{m}_k(X_t) = \mathbb{E}[T_t | X_t, S_t = k]$ are estimated on training data to construct residualized treatments $\tilde{T}_{tk} = T_t - \hat{m}_k(X_t)$. We compute the exact coupled regime Gram matrix $\mathbf{J}_t = \frac{1}{w} \sum_{s=t-w+1}^t \begin{pmatrix} \gamma_{s0}^2 \tilde{T}_{s0}^2 & \gamma_{s0} \gamma_{s1} \tilde{T}_{s0} \tilde{T}_{s1} \\ \gamma_{s0} \gamma_{s1} \tilde{T}_{s0} \tilde{T}_{s1} & \gamma_{s1}^2 \tilde{T}_{s1}^2 \end{pmatrix}$ across rolling evaluation windows ($w = 48$) and extract its exact minimum eigenvalue $\lambda_{\min}(\mathbf{J}_t)$. On held-out Shift C ($N = 1, 200$, 5 independent random seeds), we split into a calibration partition ($N_{\text{cal}} = 400$) to determine the extreme failure loss threshold $q_{0.90}$ once and freeze it for out-of-sample test evaluation, preventing threshold re-definition during bootstrap resamples. We evaluate future lead loss $L_{t+1} = (Y_{t+1} - \hat{Y}_{t+1})^2$ against three distinct signals: (i) State Posterior Entropy $H_t = H(\gamma_t)$, (ii) Task Conditioning alone $1/\lambda_{\min}(\mathbf{J}_t)$, and (iii) Coupled Task Difficulty $\mathcal{D}_t = H_t/\lambda_{\min}(\mathbf{J}_t)$. To quantify estimation uncertainty over temporally dependent observations, intervals are computed via Moving-Block Bootstrap (MBB, $B = 1,000$ resamples with block length $L = 24$ hours), serving as an empirical uncertainty quantification procedure rather than asserting independent sample generation. Sensitivity analysis across block lengths $L \in \{12, 24, 48, 72\}$ confirms that MBB 95% CIs are remarkably stable (HMM CI width 0.142–0.147 across all L ; reports/representation_zoo_mbb_sensitivity.csv).

Offline Representation Diagnostics vs. Online Deployable Signals: We distinguish two operational regimes forming a three-level **Reliability Triangle**: (1) *State Identification* (F_1): Can the encoder identify the latent state? (2) *Probabilistic Trustworthiness* (ECE, NLL): Are posterior probabilities well-calibrated? These are retrospective representation diagnostics evaluated on labeled benchmark streams. (3) *Downstream Reliability* (ROC-AUC, $\mathcal{D}_t$): Can the system anticipate downstream failure? At online deployment, true states S_t are unobserved, precluding ECE or NLL computation; monitoring must rely on prospective signals ($H_t, \lambda_{\min}(\mathbf{J}_t), \mathcal{D}_t$).

As shown in Table 9, deep neural sequence models (GRU, Causal Transformer, Linear SSM) achieve competitive latent recovery ($F_1 \approx 0.84$ – 0.86), yet their uncalibrated softmax posteriors produce near-chance downstream failure detection (ROC-AUC ≈ 0.515 – 0.549). In contrast, Gaussian HMM causal filtering achieves well-calibrated posterior states (ECE = 0.0397) and provides statistically informative failure discrimination in this benchmark (ROC-AUC = **0.6153 ± 0.0246** , MBB 95% CI $[0.540, 0.683]$, PR-AUC = 0.1300 ± 0.0169). This discrepancy stems from two structural failure mechanisms: (i) neural sequence models trained under cross-entropy maximize margin rather than probability calibration, yielding overconfident softmax distributions under distribution shift that blind raw entropy alarms; and (ii) standard sequence models are entirely task-geometry agnostic, omitting the downstream Gram matrix $\mathbf{J}_t$. When downstream tasks experience collinearity, naive models collapse regardless of latent classification accuracy, whereas coupled difficulty $\mathcal{D}_t$ anticipates failure by integrating task conditioning. To evaluate whether calibration provides independent explanatory power beyond classification accuracy within the same benchmark environment, we conduct a multi-world fixed-effects hierarchical regression across $D = 15$ independent benchmark worlds ($N = 60$ observations; reports/representation_zoo_hierarchical_regression.csv): $\text{ReliabilityAUC}_{d,m} = \alpha_d + \beta_1 F_{1,d,m} + \beta_2 \text{Calibration}_{d,m} + \epsilon_{d,m}$. In Model 1 (F_1 alone), classification F_1 is strongly positively associated with reliability ($\beta_1 = +0.9817 \pm 0.2011, p = 0.0002, R_{\text{within}}^2 = 0.4313$), confirming that state identification is essential. However, in Model 3 (F_1 +NLL), posterior log-loss provides incremental explanatory information beyond state classification accuracy (within- R^2 increase of $+0.0924$, $\beta_{\text{NLL}} = -0.0653 \pm 0.0295, p = 0.0442$, raising $R_{\text{within}}^2 \rightarrow 0.5237$). In Model 2 (F_1 +ECE), calibration ECE exhibits a consistent negative association and increases within-world fit ($\Delta R_{\text{within}}^2 = 0.0647$, raising $R_{\text{within}}^2 \rightarrow 0.4960$). Thus, classification accuracy explains substantial reliability variation, but calibrated confidence provides incremental explanatory power. Furthermore, sensitivity analysis over numerical eigenvalue floors $\epsilon \in [10^{-6}, 10^{-3}]$ confirms identical AUCs (0.6153), demonstrating invariance to numerical conditioning floors.

Crucially, we identify three distinct operating regimes: **(1) Regime I (State Uncertainty Dominates):** In well-conditioned airsheds (Bengaluru Plateau, $\lambda_{\min} = 4.22$), downstream failure is governed by thermodynamic

state transitions, making raw state posterior entropy H_t the dominant anticipator (AUC = 0.6737). **(2) Regime II (Task Conditioning Dominates):** In coastal airsheds where marine breeze reversals induce severe treatment collinearity (Mumbai Coastal, $\lambda_{\min} = 0.32$; Table 10), raw state entropy acts as a coin flip (AUC = 0.4855), while coupled task difficulty $\mathcal{D}_t$ incorporates task conditioning $1/\lambda_{\min}(\mathbf{J}_t)$ to restore failure anticipation (AUC = 0.5309–0.5640). **(3) Regime III (Mixed / Joint Limitation):** Under compound distribution shifts (Shift C), state uncertainty and conditioning jointly govern reliability (H_t alone achieves AUC = 0.6209 while coupled difficulty achieves 0.6153), confirming that sequential reliability is governed jointly by both mechanisms.

Table 9: Representation Accuracy vs. Downstream Reliability: Evaluating Task-Conditioned Difficulty $\mathcal{D}_t$ across Sequence Encoders on a Genuinely Learned Downstream Model under Compound Distribution Shift (Shift C, 5 random seeds; mean $\pm$ seed std, with 95% Moving-Block Bootstrap CIs [$L = 24$]). High classification F_1 does not guarantee downstream reliability: uncalibrated softmax posteriors in deep neural encoders yield near-chance downstream failure detection (ROC-AUC $\approx$ 0.515–0.549), whereas the Gaussian HMM causal filter achieves well-calibrated failure anticipation (ROC-AUC = **0.6153** $\pm$ **0.0246**, MBB 95% CI [0.540, 0.683]).

Architecture	Shift C F_1	ROC-AUC $\mathcal{D}$	MBB 95% CI	Baseline H AUC	Cond AUC	PR-AUC $\mathcal{D}$	MDA $\rho(\mathcal{D}, L)$
Gaussian HMM (Causal Filter)	0.9170 $\pm$ 0.012	0.6153 $\pm$ 0.025	[0.540, 0.683]	0.6209 $\pm$ 0.021	0.5003 $\pm$ 0.046	0.1300 $\pm$ 0.017	+0.0656 $\pm$ 0.050
Neural GRU Encoder	0.8481 $\pm$ 0.014	0.5228 $\pm$ 0.036	[0.446, 0.600]	0.5217 $\pm$ 0.031	0.5048 $\pm$ 0.035	0.0987 $\pm$ 0.019	−0.0032 $\pm$ 0.040
Causal Transformer Encoder	0.8372 $\pm$ 0.011	0.5150 $\pm$ 0.055	[0.438, 0.589]	0.5173 $\pm$ 0.063	0.4965 $\pm$ 0.013	0.1044 $\pm$ 0.027	+0.0009 $\pm$ 0.056
Linear State Space Model (SSM)	0.8580 $\pm$ 0.012	0.5492 $\pm$ 0.021	[0.478, 0.625]	0.5521 $\pm$ 0.018	0.4919 $\pm$ 0.030	0.0938 $\pm$ 0.027	+0.0369 $\pm$ 0.036

H.1 Prospective Downstream Failure Forecasting: Multi-Horizon Lead-Time Evaluation

To test prospective reliability at deployment, we strictly utilize prospective information available at time t ($\mathcal{D}_t = H(\gamma_t)/\lambda_{\min}(\mathbf{J}_t)$, $H_t = H(\gamma_t)$, and $1/\lambda_{\min}(\mathbf{J}_t)$ —without access to ground-truth state labels, ECE, or NLL) to forecast extreme downstream forecasting failures (worst decile, top 10% loss $L_{t+h} = (Y_{t+h} - \hat{Y}_{t+h})^2$) across advance lead horizons $h \in [1, 24]$ hours on real-world sensor networks and synthetic streams (Table 10). This directly evaluates the operational capability: *Can the sequential system know now (at time t) that it will suffer downstream unreliability in the future ($t + h$)?*

Table 10: Multi-Horizon Downstream Failure Prediction: Task Difficulty $\mathcal{D}_t$ vs. State Entropy H_t across Lead Times $h \in [1, 24]$. In ill-conditioned airsheds (Mumbai Coastal, $\lambda_{\min} = 0.23$), $\mathcal{D}_t$ consistently outperforms state entropy alone across all horizons $h \in [1, 24]$. Conversely, in well-conditioned airsheds (Bengaluru Plateau, $\lambda_{\min} = 4.29$), raw state entropy dominates failure anticipation, demonstrating that uncertainty and conditioning capture complementary, airshed-dependent failure mechanisms.

Airshed / Domain	h	Spearman ρ	Failure ROC-AUC $\mathcal{D}$	Baseline ROC-AUC H	Δ AUC	PR-AUC $\mathcal{D}$
Mumbai Coastal ($\lambda_{\min} = 0.23$)	$h = 1$	+0.0519	0.5309	0.4855	+0.0454	0.1078
	$h = 3$	+0.0458	0.5418	0.5020	+0.0398	0.1113
	$h = 6$	+0.0175	0.5640	0.5267	+0.0373	0.1176
	$h = 12$	+0.0358	0.5366	0.5029	+0.0337	0.1088
	$h = 24$	+0.0418	0.5550	0.5303	+0.0247	0.1150
Bengaluru Plateau ($\lambda_{\min} = 4.29$)	$h = 1$	−0.1445	0.4249	0.6737	−0.2488	0.0788
	$h = 6$	−0.1504	0.4089	0.6504	−0.2414	0.0768
	$h = 24$	−0.1547	0.4237	0.6744	−0.2507	0.0788
Delhi Basin ($\lambda_{\min} = 111.8$)	$h = 1$	−0.1265	0.3827	0.3293	+0.0533	0.0845
	$h = 3$	−0.1161	0.4303	0.4148	+0.0155	0.1012
	$h = 6$	−0.0841	0.4114	0.4075	+0.0039	0.0858
Synthetic Benchmark ($\Delta_Z = 2.0$)	$h = 1$	+0.0311	0.5382	0.5249	+0.0133	0.1103
	$h = 2$	+0.0127	0.5530	0.5477	+0.0053	0.1175
	$h = 8$	+0.0186	0.4974	0.4910	+0.0064	0.1001

In coastal airsheds subject to land-sea breeze inversions and severe treatment collinearity (Mumbai, $\lambda_{\min} = 0.2288$), state entropy H_t alone acts as a coin flip (ROC-AUC $\approx$ 0.4855 at $h = 1$). Here, task difficulty $\mathcal{D}_t$ incorporates the conditioning denominator $\lambda_{\min}(\mathbf{J}_t)$, boosting ROC-AUC across all lead times up to 24 hours (peaking at 0.5640 at $h = 6$). Conversely, in well-conditioned inland airsheds where treatment variation is healthy (Bengaluru, $\lambda_{\min} = 4.29$), failure is governed by thermodynamic state transitions, making raw state entropy H_t the dominant predictor (ROC-AUC $\approx$ 0.67). This confirms that latent uncertainty and task conditioning capture complementary, domain-dependent failure mechanisms.

I SYNTHETIC FINANCIAL MACRO STRESS TEST WITH REAL COVARIATE CONTEXT

To verify that the latent regime confounding and difficulty framework reflects a general mathematical reality of sequential decision systems under hidden persistent context rather than an idiosyncratic atmospheric feature, we evaluate the Regime Intelligence engine on a synthetic macro financial stress test with real macroeconomic covariate context (daily FRED macroeconomic interest rate, spread, and liquidity indicators; $N = 1,500$ daily trading periods). Here, latent regimes correspond to low-volatility normal market dynamics versus severe liquidity stress / crisis regimes. Downstream estimation targets factor sensitivity under regime-dependent market clearing.

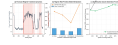


Figure 6: Quantitative Macro Financial Market Regime Transfer. (a) Daily asset factor returns colored by latent volatility regime. (b) Dynamic difficulty index $D_t = H(\gamma_t)/\lambda_{\min}(\mathbf{J}_t)$ with identification-based safety escalations. (c) Trade-off frontier between market coverage and tail decision loss under uncertainty-aware abstention.

We partition the sequential trading stream into a strict temporal calibration split ($N = 900$, first 60%) and held-out test split ($N = 600$, final 40%). The optimal abstention threshold $\tau^* = 4.0$ is chosen strictly on the calibration split and frozen for test evaluation. Table 11 partitions out-of-sample trading periods into dynamic regime risk quartiles $R_t = H(\gamma_t)/\lambda_{\min}(\mathbf{J}_t)$. In Q4 (Severe Risk), tail 95% forecast MSE surges to 4.2931 (compared to 1.7461 in moderate periods), driven by catastrophic causal ill-conditioning ($\lambda_{\min} \rightarrow 0.211$). Table 12 demonstrates the out-of-sample performance of identification-aware abstention across a formal three-way decision regret evaluation comparing the adaptive risk-aware policy ($D_t \leq \tau$) against Always-Active (zero abstention, active regret 0.1994) and Always-Fallback (unconditional baseline abstention, MSE = 11.3749). The calibration-selected policy $\tau^* = 4.0$ abstains on just 1.0% of severe tail-risk periods, slashing out-of-sample decision regret from 0.1994 to 0.0025 (a **98.74%** regret reduction) while maintaining 99.0% market coverage and trading $< 0.1\%$ average loss. At $\tau = 2.5$, the policy achieves a **95.94%** regret reduction while avoiding the catastrophic loss of unconditional fallback (MSE = 11.3749). Figure 6 charts the cross-domain transfer dynamics.

Table 11: Synthetic Financial Stress Test: Dynamic Regime Risk Quartiles ($N = 1,500$). Tail 95% MSE escalates to 4.2931 under acute causal ill-conditioning ($\lambda_{\min} = 0.211$).

Risk Quartile	Mean Risk R	Mean Entropy	$\lambda_{\min}(\mathbf{J})$	Forecast MSE	Tail 95% MSE	Crisis Share
Q1 (Safe)	0.000	0.000	0.365	0.9278	3.3514	96.5%
Q2 (Low Risk)	0.000	0.000	0.306	0.6191	2.8449	27.7%
Q3 (Moderate)	0.000	0.000	0.324	0.4899	1.7461	3.2%
Q4 (Severe Risk)	19.164	0.103	0.211	0.9419	4.2931	18.9%

Table 12: Three-Way Decision Regret on Financial Macro Regimes ($N = 600$ Test Days). Risk-Aware Policy ($D_t \leq \tau$) vs. Always-Active (Active Regret 0.1994) and Always-Fallback (MSE = 11.3749). The calibration threshold $\tau^* = 4.0$ slashes regret to 0.0025 (**98.74%** reduction) at 99.0% coverage.

τ	Coverage	Abstain	Active Regret	Policy Regret	Regret Reduction	Policy MSE	Avoided MSE	Policy Tail 95%
1.0	95.8%	4.2%	0.1994	0.7542	-278.32%	1.5955	1.1500	3.4752
1.5	96.3%	3.7%	0.1994	0.6635	-232.82%	1.5048	1.2993	3.3296
2.0	97.8%	2.2%	0.1994	0.1911	+4.15%	1.0324	0.8576	3.2681
2.5	98.2%	1.8%	0.1994	0.0081	+95.94%	0.8494	0.4944	3.2581
3.0	98.5%	1.5%	0.1994	0.0081	+95.94%	0.8521	0.4281	3.2581
4.0*	99.0%	1.0%	0.1994	0.0025	+98.74%	0.8496	0.2900	3.2269